High-efficiency Heating Circulator

Calio Pro Plus

Type Series Booklet

Legal information/Copyright

Type Series Booklet Calio Pro Plus

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Contents

Heating

Variable Speed Heating Circulators

Calio Pro Plus

Main applications

- Heating, ventilation, air-conditioning, cooling and circulation systems
- One-pipe systems and two-pipe systems
- Underfloor heating systems
- Boiler circuits or primary circuits
- Storage tank circuits
- Solar power systems
- Heat pumps

Fluids handled

- Heating water to VDI 2035
- Higher-viscosity fluids (water/glycol mixture up to a mixing ratio of 1:1)

Operating data

Table 1: Operating properties

Characteristic		Value	
		$P_1 \leq 400$ W	$P_1 > 400$ W
Flow rate	Q [m³/h]	≤ 69	
	Q [l/s]	≤ 19,2	
Head	H [m]	≤ 18	
Fluid temperature	T [°C]	≥ -10	
		≤ +110	
Ambient temperature	T [°C]	≥ 0	
		≤ +40	
Relative humidity	rF [%]	≤ 95	
Operating pressure	p [bar]	≤ 16	
Pressure class	PN [bar]	6/10/16	
Average sound pressure level	[dB (A)]	< 40	< 55
Screw-ended connection	G	1 1/2 - 2	
Flanged connection	DN	32 - 100	

Design details

Design

- Maintenance-free high-efficiency wet rotor pump (glandless)

Drive

- High-efficiency permanent magnet synchronous motor, brushless, self-cooling, with continuously variable differential pressure control
- 1~230 V AC +/- 10%
- Frequency 50 Hz/60 Hz
- Enclosure IPX4D
- Thermal class F
- Temperature class TF 110
- Energy efficiency index EEI ≤ 0.18[1)]
- Interference emissions EN 55014-1, EN 61000-3-2, EN 61000-3-3

$P_1 \leq 400$ W:

- Interference immunity EN 55014-2

$P_1 > 400$ W:

- Interference immunity EN 55014-2EN 61000-6-2

Bearings

- Product-lubricated special plain bearing

Connections

- Screw-ended or flanged

Operating modes

- Constant-pressure control
- Proportional-pressure control
- Dynamic Control
- Constant speed (open-loop control)
- Constant-flow-rate control

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1 The reference value for the most efficient of circulators is EEI ≤ 0.20.

- Constant-temperature control
- Temperature-governed differential pressure control (can only be activated via the KSB FlowManager app or KSB ServiceTool)
- Differential-temperature control (can only be activated via the KSB FlowManager app or KSB ServiceTool)[2)]

Automatic functions

- Continuously variable speed adjustment depending on the mode of operation
- Dual-pump operation
- Setback operation
- Calorimeter [2)]
- Deblocking function
- Self-venting function of the pump casing
- Functional check run
- Soft start
- Full motor protection with integrated trip electronics

Manual functions

- Remote ON/OFF
- Setting the operating mode
- Setting the discharge head setpoint
- Setting the flow rate setpoint
- Setting the temperature setpoint
- Setting the speed
- Rotor space venting function
- Locking the control panel
- 0/2 - 10 V with external differential pressure/speed setpoint
- 0 - 10 V as input of the actual value of the temperature or actual value of the differential pressure

Signalling functions and display functions

- Periodically alternating display of flow rate, head and electrical input power
- Operating condition and error codes shown on the display, in the KSB FlowManager app or in the KSB ServiceTool
- Configurable general fault message and "in operation" message (volt-free changeover contacts)
- Serial digital Modbus RTU interface
- Bluetooth interface for the KSB FlowManager app or KSB ServiceTool
- Service interface for KSB ServiceTool

Designation

Example: Calio Pro Plus 25-40

Table 2: Designation key

Code	Description	
Calio Pro Plus	Type series	
	_[3)]	Single pump
	Z	Twin pump
25	Connection	
	25	G 1 1/2
	30	G 2
	32	DN 32
	40	DN 40
	50	DN 50
	65	DN 65
	80	DN 80
	100	DN 100
40	Head H[4)] [m]	
	40	Head × 10 Example: 4 m × 10 = 40

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2 One external temperature sensor is required as a minimum.
3 Blank
4 At flow rate Q = 0 m³/h

Materials

Table 3: Overview of available materials

Part No.	Description	Material
102	Volute casing	Grey cast iron with cathodic electrocoating (EN-GJL-200)
210	Shaft	Stainless steel 1.4034
230	Impeller	Glass fibre reinforced plastics (P1 ≤ 400 W: PSU-GF30, P1 > 400 W: PPS-GF40)
310	Bearings	Ceramics / carbon
689	Thermal insulation shells	Polypropylene
817	Can	Glass fibre reinforced plastic (PPS-GF40)

Casing parts which are in contact with the atmosphere and with the fluid handled are free from paint-wetting impairment materials.

Product benefits

- **Highest energy efficiency with Dynamic Control**
  - Patented Dynamic Control for up to 40 percent additional energy savings as compared with proportional-pressure control
  - Practically rules out undersupply in changing system conditions
  - Optimised efficiencies exceeding the requirements of the energy efficiency regulations
- **Easy to install and commission**
  - All operating modes, functions and setpoints in metres are easy and fast to adjust at the pump or via the Bluetooth interface.
  - Quick and straightforward commissioning with the FlowManager app
  - With the FlowManager app all electrical connections can easily and flexibly be configured for the specific application.
  - In the FlowManager app all pump information is quickly accessible via the Favourites menu. Process values are clearly illustrated in a diagram.
- **Universal use**
  - One pump for three pressure classes: PN 6/PN 10/PN 16
  - New and enhanced characteristic curves offer a larger operating range.
  - Flexible pump functions, such as an operating point display, integrated dual-pump operation and heat meter
- **High operating reliability**
  - High operating reliability thanks to built-in pump and motor protection equipment
  - A maintenance-free motor with a carbon fibre reinforced plastic can and an integrated filtering concept provide protection against contamination with magnetite, for example.
- **Connectivity**
  - Modbus RTU interface integrated as standard
  - Flexible connectivity by plug-in modules for all common building management systems
  - Two parameterisable relays for operation and fault messages
  - Configurable digital input and adjustable analog input
- **Environmentally friendly**
  - Maximum energy efficiency
  - Energy savings meter
  - Life Cycle Impact Assessment (LCIA) based on ISO 14040 / 14044

Product information

Product information as per Regulation No. 1907/2006 (REACH)

For information as per European chemicals regulation (EC) No. 1907/2006 (REACH) see https://www.ksb.com/en-global/company/corporate-responsibility/reach.

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Data access and use in accordance with Regulation (EU) 2023/2854 (EU Data Act)

Table 4: Device data recorded

Data type	Format	Volume	Continuous generation / real-time generation	Data storage		Duration of retention
(Article 3, 2a)	**(Article 3, 2a)**	**(Article 3, 2a)**	**(Article 3, 2b)**	**(Article 3, 2c)**		**(Article 3, 2c)**
				Device	**Remote server**	
Stator winding temperatures (NTC)	FLOAT32	4 bytes	Yes	Yes	No	Instantaneous value
SPM temperature (power electronics)	FLOAT32	4 bytes	Yes	Yes	No	Instantaneous value
Current measurement (drive end)	FLOAT32	4 bytes	Yes	Yes	No	Instantaneous value
DC link voltage	FLOAT32	4 bytes	Yes	Yes	No	Instantaneous value
Analog input	FLOAT32	4 bytes	Yes	Yes	No	Instantaneous value
Digital input	BOOL	1 byte	Yes	Yes	No	Instantaneous value
Pt1000 resistance thermometer	FLOAT32	4 bytes	Yes	Yes	No	Instantaneous value
Operating hours Pump	UINT32	4 bytes	Yes	Yes	-	Unlimited
Control board temperature	FLOAT32	4 bytes	Yes	Yes	-	Instantaneous value
Mains voltage	FLOAT32	4 bytes	Yes	Yes	-	Instantaneous value

Data access (Art. 3, 2d)

Authorised users can access the collected data in the following ways:

- Data stored on the device
  - **Access:** The data stored on the device can be accessed using the KSB ServiceTool.
  - **Interface**: The data stored on the device can be read out via an optional field bus interface.
- When an optional field bus interface is used, data may be stored on a **remote server** provided by the customer.
  - **Access:** The customer is responsible for managing, securing and updating the access data.
  - **Technical details:** The customer is responsible for access rights, data storage and data deletion.
  - **Data format**: Exported data can be processed in commonly used tools (e.g. Excel, ERP systems).
- When an optional field bus interface is used, no data is stored on a **KSB remote server**.

The requirements regarding the information obligation under the EU Data Act for related services such as the KSB ServiceTool and the KSB FlowManager are set out in a separate document. Only the data of the connected product itself is shown here.

Available software/apps

KSB ServiceTool

PACTware™

This software can be used to change the settings for the pump set and carry out a firmware update.

https://www.ksb.com/en-gb/software-and-know-how/operational-tools/ksb-servicetool

KSB FlowManager 2.0

This app can be used to change the settings for the pump set and carry out a firmware update.

Download on the App Store

GET IT ON Google Play

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Selection information

Minimum inlet pressure

The minimum inlet pressure p_{min} at the pump suction nozzle serves to avoid cavitation noises at the indicated fluid temperature $T_{max.}$.

The values indicated are gauge pressures relative to the ambient pressure and are valid for altitudes of up to 300 m above seal level. For installation at altitudes > 300 m, an allowance of 0.01 bar / 100 m must be added.

Table 5: Minimum inlet pressure p_{min} specified for the fluid temperature $T_{max.}$

Fluid temperature	Minimum inlet pressure
[°C]	[bar]
≤ 80	1,0
81 to 95	1,5
96 to 110	2,5

Permissible fluid temperature

Table 6: Temperature limits of the fluid handled

Permissible fluid temperature	Value
Maximum	+110 °C
Minimum	-10 °C

Permissible ambient temperature

Table 7: Permissible ambient temperatures

Permissible ambient temperature	Value
Maximum	+40 °C
Minimum	0 °C

Description of the Modbus interface

Table 8: Technical data of the Modbus interface

Characteristic	Value
Terminal cross-section	▪ 1.5 mm^2 for rigid or flexible conductors ▪ 1.0 mm^2 with wire end sleeve without plastic sleeve ▪ 0.5 mm^2 with wire end sleeve and plastic sleeve
Interface	RS485 (TIA-485A)
Bus connection	0.5 mm^2 Modbus network cable twisted in pairs and shielded
Cable length	▪ ≤ 1000 m ▪ Stub line impermissible ▪ For electric cable lengths > 30 m take suitable measures to ensure overvoltage protection.
Impedance	120 Ω (cable type B to TIA 485-A)
Data rates	4800, 9600, 38400, 57600, 115200 baud (19200 baud = factory setting)
Protocol	Modbus RTU standard
Data format	▪ 8 data bits ▪ Parity EVEN / ODD / NONE ▪ 1 stop bit
Modbus address	ID #1 to #247 selectable (ID #17 = factory setting)

Further description see operating manual of the pump set.

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Description of the Dynamic Control function

The dynamic control (2) system detects when the selected control curve (3) is higher than the minimum characteristic curve[5] (4). The control system shifts the control curve downward, and power input is reduced automatically. To ensure sufficient supply the pump set switches to a higher control curve when the minimum characteristic curve is reached. The energy input is reduced (1) without any negative impact on the supply of the building.
The pump set is operated in an optimised way, even if the system characteristic curve is unknown; the noise at the thermostatic valves is reduced.

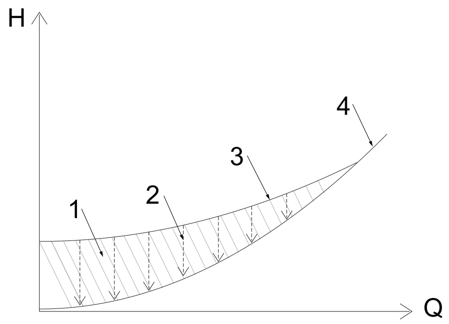

Fig. 1: Principle of Dynamic Control

1	Excess energy input	3	Control curve
2	Dynamic Control	4	Minimum characteristic curve

Description of the characteristic curve

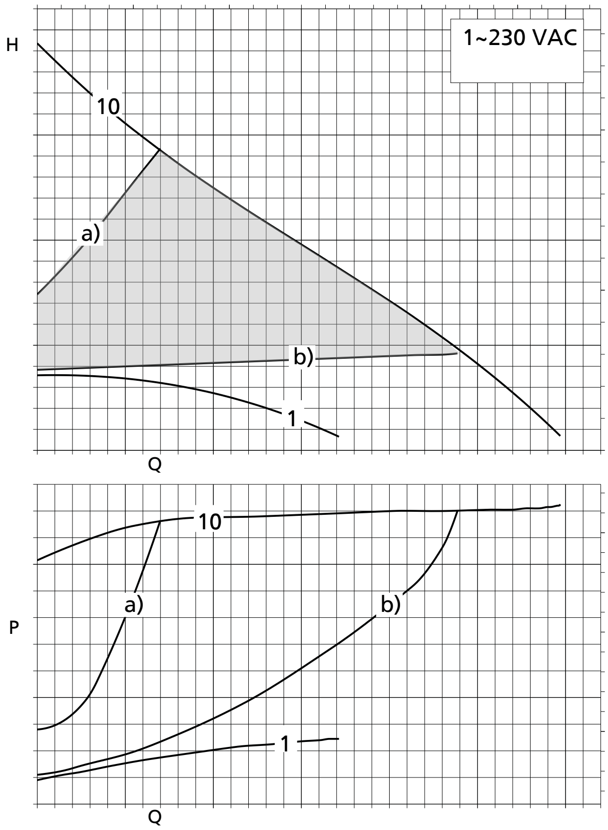

Fig. 2: Selection example

1	Minimum fixed speed operation
10	Maximum fixed speed operation
	Control range
a)	Control curve, maximum head
b)	Control curve, minimum head

The characteristic curve can be adjusted between a) and b) in increments of 0.1 m. This adjustment can be made with the control buttons.

5 Characteristic curve at fully open thermostatic valves

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Technical data

Calio Pro Plus

Table 9: Technical data

Size	Connection		PN	n		P_1	I_N	Mat. No.	[kg]
	Piping	Pump	[bar]	Min.	Max.	[W]	1~230 V AC, 50 Hz/60 Hz		
				[rpm]	[rpm]		[A]		
25-40	R 1[6]	G 1 1/2	6/10/16	1000	3000	2,5 - 70	0,10 - 0,74	29135464	5.5
25-60	R 1[6]	G 1 1/2	6/10/16	1000	3600	2,5 - 110	0,10 - 1,04	29135465	5.5
25-80	R 1[6]	G 1 1/2	6/10/16	1000	4200	2,5 - 165	0,10 - 1,20	29135466	5.5
25-100	R 1[6]	G 1 1/2	6/10/16	1000	4600	2,5 - 225	0,10 - 1,23	29135467	5.8
25-120	R 1[6]	G 1 1/2	6/10/16	1000	5000	2,5 - 400	0,10 - 2,04	29135468	6.2
30-40	R 1 1/4[6]	G 2	6/10/16	1000	3000	2,5 - 90	0,10 - 0,82	29135469	5.7
30-60	R 1 1/4[6]	G 2	6/10/16	1000	3600	2,5 - 120	0,10 - 1,10	29135470	5.7
30-80	R 1 1/4[6]	G 2	6/10/16	1000	4100	2,5 - 175	0,10 - 1,11	29135471	5.7
30-100	R 1 1/4[6]	G 2	6/10/16	1000	4600	2,5 - 225	0,10 - 1,12	29135472	6
30-120	R 1 1/4[6]	G 2	6/10/16	1000	4000	2,5 - 400	0,10 - 1,93	29135473	6.6
30-140	R 1 1/4[6]	G 2	6/10/16	1000	4300	2,5 - 400	0,10 - 1,93	29135474	6.6
30-160	R 1 1/4[6]	G 2	6/10/16	1000	4600	2,5 - 400	0,10 - 1,93	29135475	6.6
32-40	DN 32	DN 32	6/10/16	1000	3000	2,5 - 90	0,10 - 0,82	29135476	8.9
32-60	DN 32	DN 32	6/10/16	1000	3600	2,5 - 120	0,10 - 1,10	29135477	8.9
32-80	DN 32	DN 32	6/10/16	1000	4100	2,5 - 175	0,10 - 1,11	29135478	8.9
32-100	DN 32	DN 32	6/10/16	1000	4600	2,5 - 225	0,10 - 1,12	29135479	9.2
32-120	DN 32	DN 32	6/10/16	1000	4000	2,5 - 400	0,10 - 1,93	29135480	9.8
32-140	DN 32	DN 32	6/10/16	1000	4300	2,5 - 400	0,10 - 1,93	29135481	9.8
32-160	DN 32	DN 32	6/10/16	1000	4600	2,5 - 400	0,10 - 1,93	29135482	9.8
40-40	DN 40	DN 40	6/10/16	1000	3200	2,5 - 125	0,10 - 1,09	29135483	8.9
40-60	DN 40	DN 40	6/10/16	1000	3200	2,5 - 225	0,10 - 1,32	29135484	9.3
40-80	DN 40	DN 40	6/10/16	1000	3700	2,5 - 325	0,10 - 1,56	29135485	11.6
40-100	DN 40	DN 40	6/10/16	1000	4100	2,5 - 390	0,10 - 1,88	29135486	11.6
40-120	DN 40	DN 40	6/10/16	1000	3000	2,5 - 565	0,35 - 2,64	29135487	19.5
40-150	DN 40	DN 40	6/10/16	1000	3300	2,5 - 810	0,35 - 3,77	29135488	19.5
40-180	DN 40	DN 40	6/10/16	1000	3600	2,5 - 915	0,35 - 4,26	29135489	19.5
50-40	DN 50	DN 50	6/10/16	1000	2900	2,5 - 215	0,10 - 1,34	29135490	10.5
50-60	DN 50	DN 50	6/10/16	1000	3300	2,5 - 305	0,10 - 1,45	29135491	13
50-80	DN 50	DN 50	6/10/16	1000	3800	2,5 - 390	0,10 - 1,89	29135492	13
50-100	DN 50	DN 50	6/10/16	1000	2700	2,5 - 735	0,35 - 3,43	29135493	20.4
50-120	DN 50	DN 50	6/10/16	1000	3000	2,5 - 895	0,35 - 4,18	29135494	20.4
50-150	DN 50	DN 50	6/10/16	1000	3300	2,5 - 1365	0,35 - 6,42	29135495	22.5
50-180	DN 50	DN 50	6/10/16	1000	3600	2,5 - 1780	0,35 - 8,40	29135496	22.5
65-40	DN 65	DN 65	6/10/16	1000	2700	2,5 - 225	0,10 - 1,31	29135497	17.7
65-60	DN 65	DN 65	6/10/16	1000	3100	2,5 - 375	0,10 - 1,92	29135498	17.7
65-80	DN 65	DN 65	6/10/16	1000	2800	2,5 - 545	0,35 - 2,55	29135499	27.7
65-100	DN 65	DN 65	6/10/16	1000	3100	2,5 - 695	0,35 - 3,25	29135500	27.7
65-120	DN 65	DN 65	6/10/16	1000	3400	2,5 - 835	0,35 - 3,91	29135501	27.7
80-40	DN 80	DN 80	6	1000	2200	2,5 - 540	0,35 - 2,55	29135503	29.5
80-40	DN 80	DN 80	10/16	1000	2200	2,5 - 540	0,35 - 2,55	29135504	29.5
80-60	DN 80	DN 80	6	1000	2400	2,5 - 715	0,35 - 3,32	29135505	29.5
80-60	DN 80	DN 80	10/16	1000	2400	2,5 - 715	0,35 - 3,32	29135506	29.5
80-80	DN 80	DN 80	6	1000	2700	2,5 - 1020	0,35 - 4,81	29135507	31.6
80-80	DN 80	DN 80	10/16	1000	2700	2,5 - 1020	0,35 - 4,81	29135508	31.6
80-100	DN 80	DN 80	6	1000	2900	2,5 - 1345	0,35 - 6,33	29135509	31.6
80-100	DN 80	DN 80	10/16	1000	2900	2,5 - 1345	0,35 - 6,33	29135510	31.6

6 Connection using pump pipe unions (accessories)

Size	Connection		PN	n		P_1	I_N	Mat. No.	[kg]
	Piping	Pump	[bar]	Min.	Max.	[W]	1~230 V AC, 50 Hz/60 Hz		
				[rpm]	[rpm]		[A]		
80-120	DN 80	DN 80	6	1000	3100	2,5 - 1555	0,35 - 7,31	29135511	31.6
80-120	DN 80	DN 80	10/16	1000	3100	2,5 - 1555	0,35 - 7,31	29135512	31.6
80-160	DN 80	DN 80	6	1000	3500	2,5 - 1625	0,35 - 7,55	29135513	31.6
80-160	DN 80	DN 80	10/16	1000	3500	2,5 - 1625	0,35 - 7,55	29135514	31.6
100-40	DN 100	DN 100	6	1000	1800	2,5 - 440	0,35 - 2,05	29135515	38.6
100-40	DN 100	DN 100	10/16	1000	1800	2,5 - 440	0,35 - 2,05	29135516	38.6
100-60	DN 100	DN 100	6	1000	2100	2,5 - 630	0,35 - 2,95	29135517	38.6
100-60	DN 100	DN 100	10/16	1000	2100	2,5 - 630	0,35 - 2,95	29135518	38.6
100-80	DN 100	DN 100	6	1000	2400	2,5 - 915	0,35 - 4,22	29135519	40.7
100-80	DN 100	DN 100	10/16	1000	2400	2,5 - 915	0,35 - 4,22	29135520	40.7
100-100	DN 100	DN 100	6	1000	2700	2,5 - 1390	0,35 - 6,52	29135521	40.7
100-100	DN 100	DN 100	10/16	1000	2700	2,5 - 1390	0,35 - 6,52	29135522	40.7
100-120	DN 100	DN 100	6	1000	3000	2,5 - 1395	0,35 - 6,53	29135523	40.7
100-120	DN 100	DN 100	10/16	1000	3000	2,5 - 1395	0,35 - 6,53	29135524	40.7

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Selection chart

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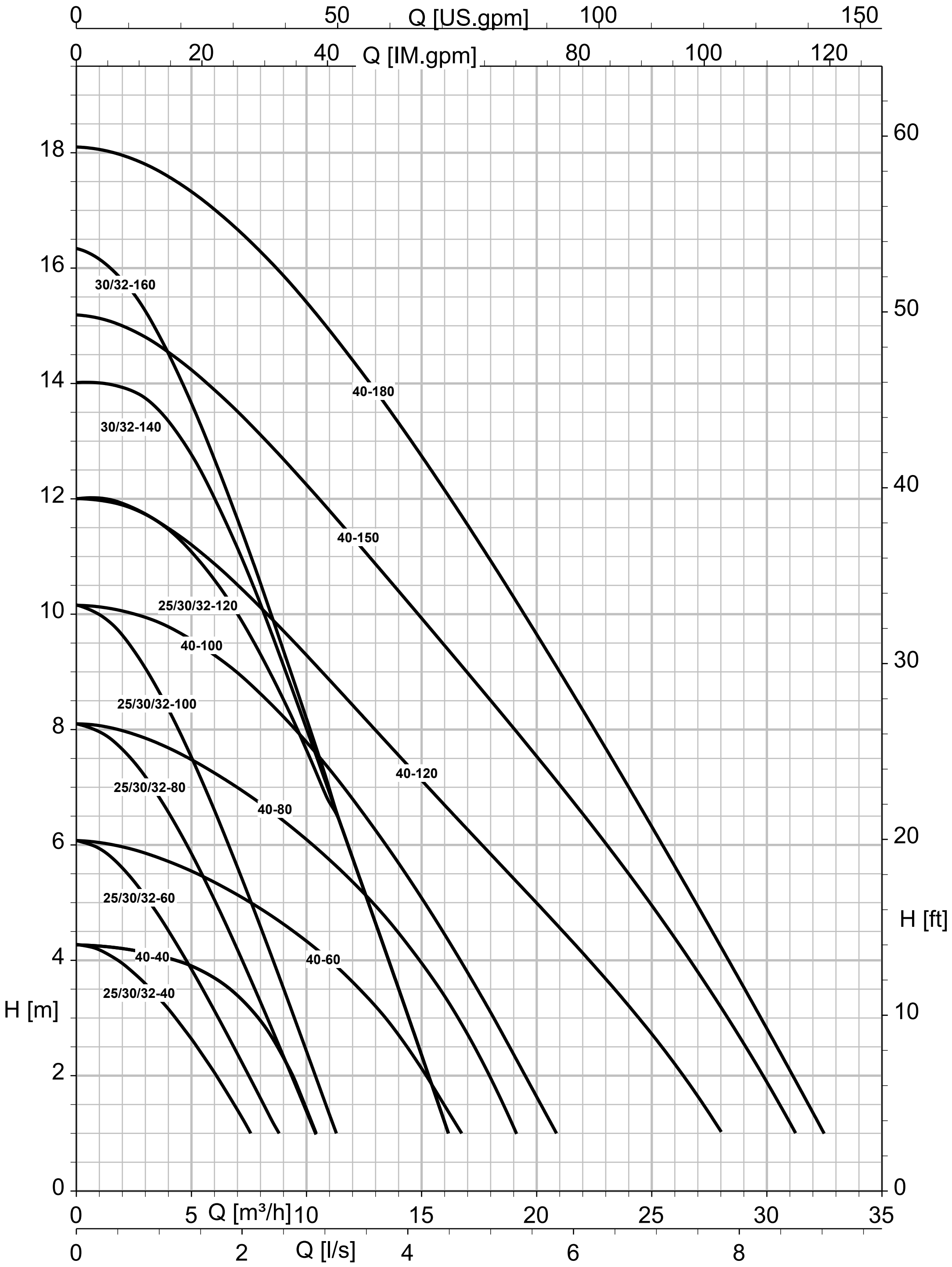

Fig. 3: Selection chart (connections 25-..., 30-..., 32-..., 40-...)

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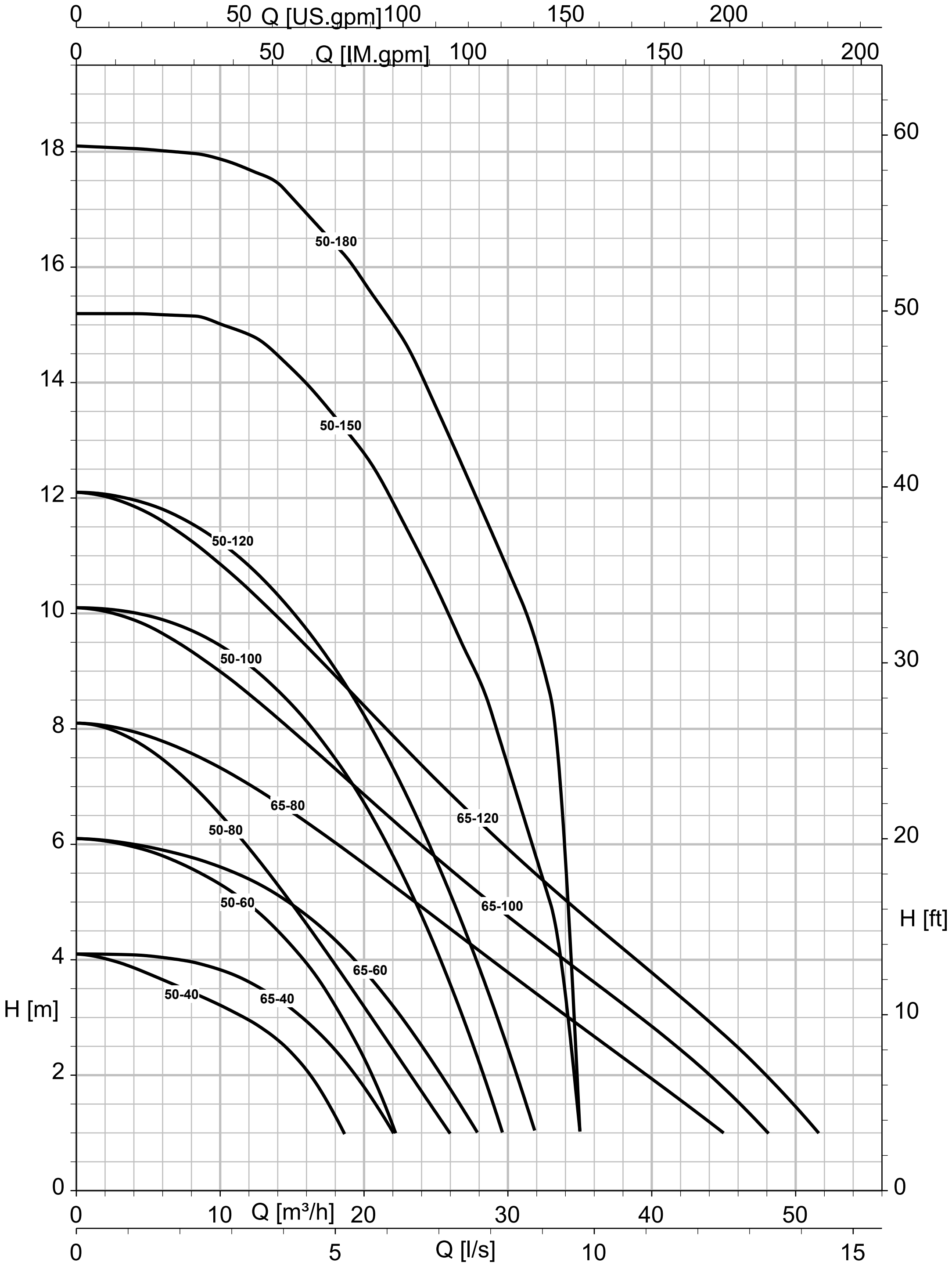

Fig. 4: Selection chart (connections 50-..., 65-...)

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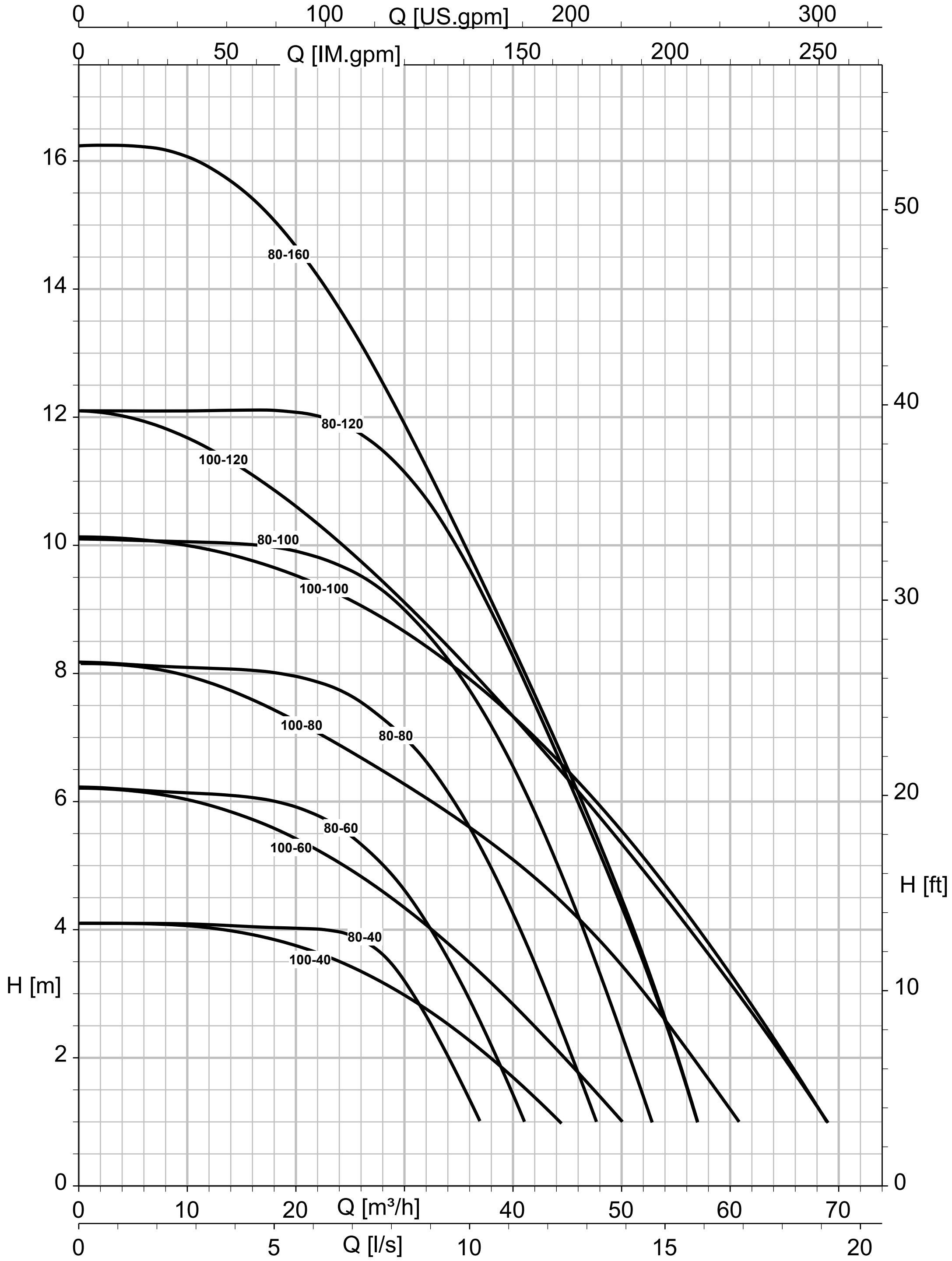

Fig. 5: Selection chart (connections 80-..., 100-...)

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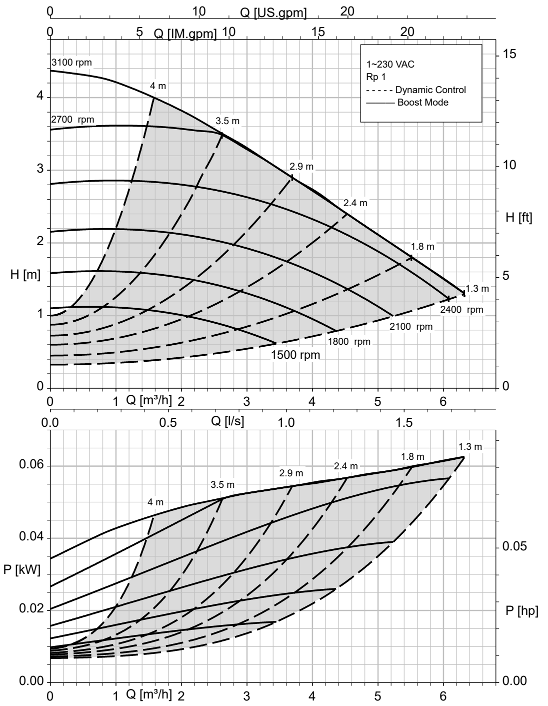
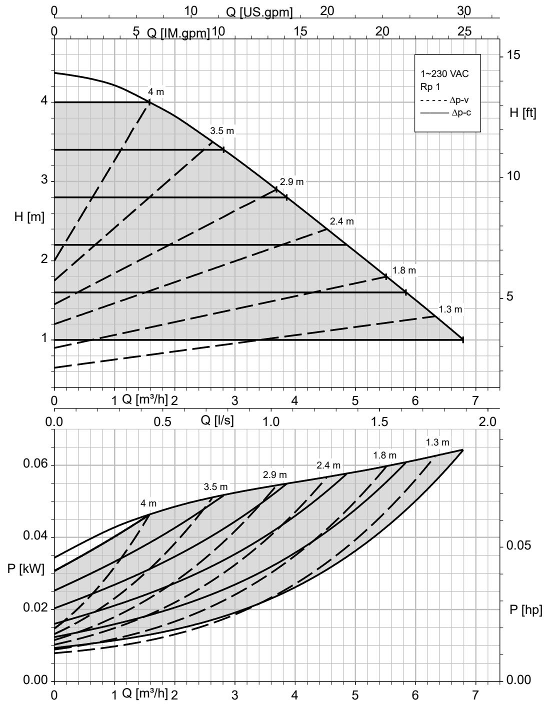
Characteristic curves

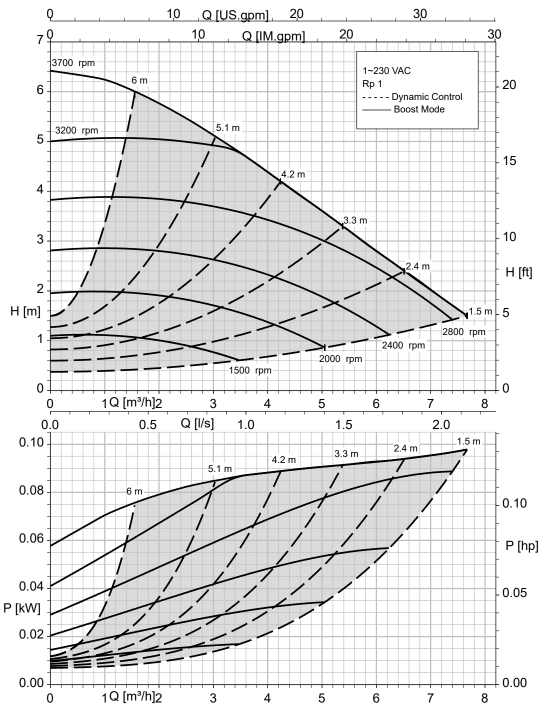
Calio Pro Plus 25-40 Open-loop Control, Dynamic Control



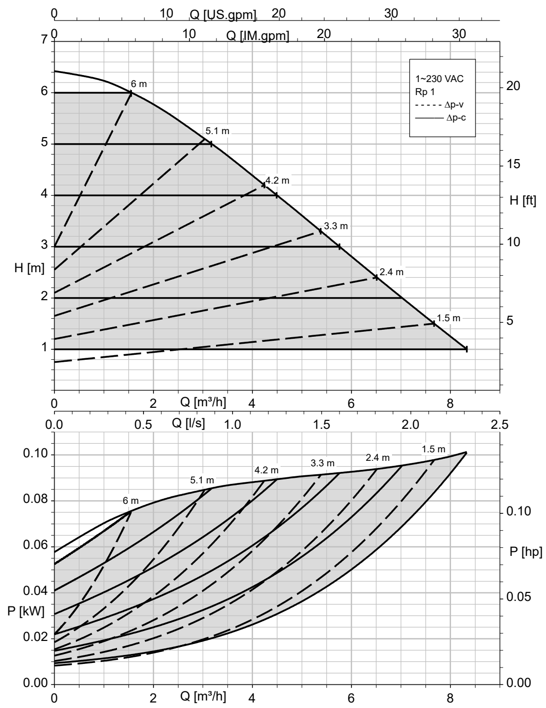
Calio Pro Plus 25-40 Δpv, Δpc



Calio Pro Plus 25-60 Open-loop Control, Dynamic Control



Calio Pro Plus 25-60 Δpv, Δpc



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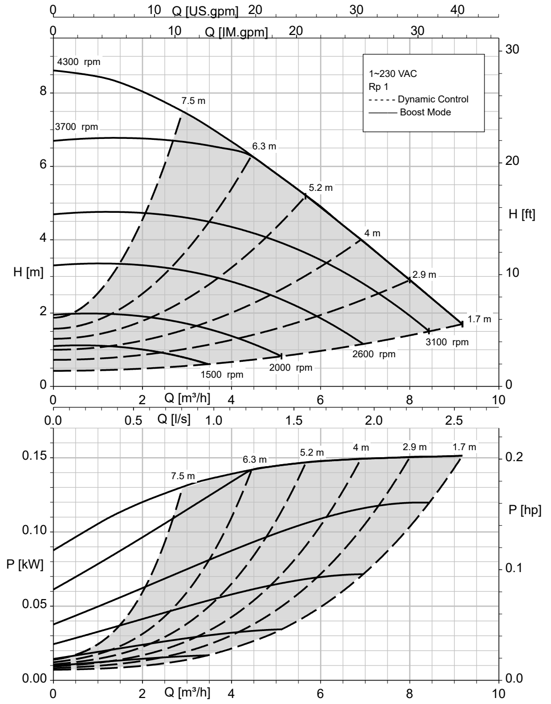
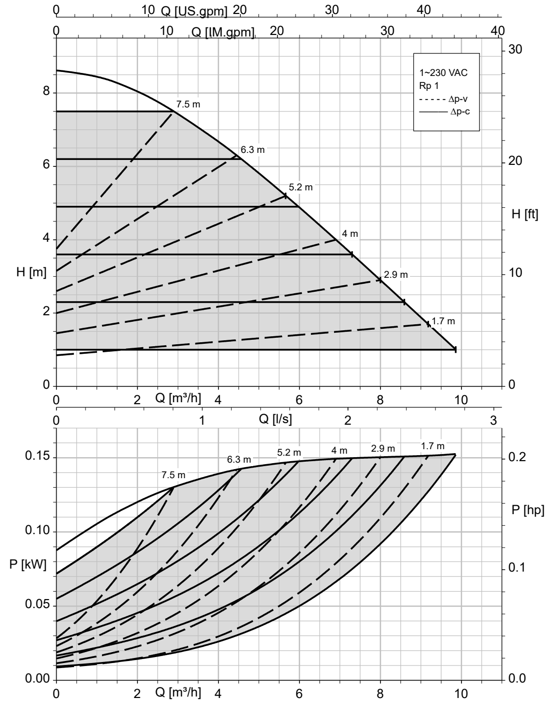
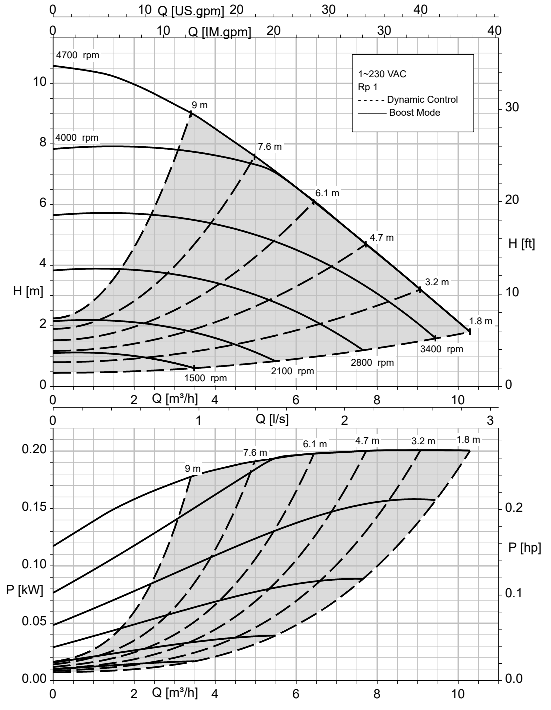
Calio Pro Plus 25-80 Open-loop Control, Dynamic Control



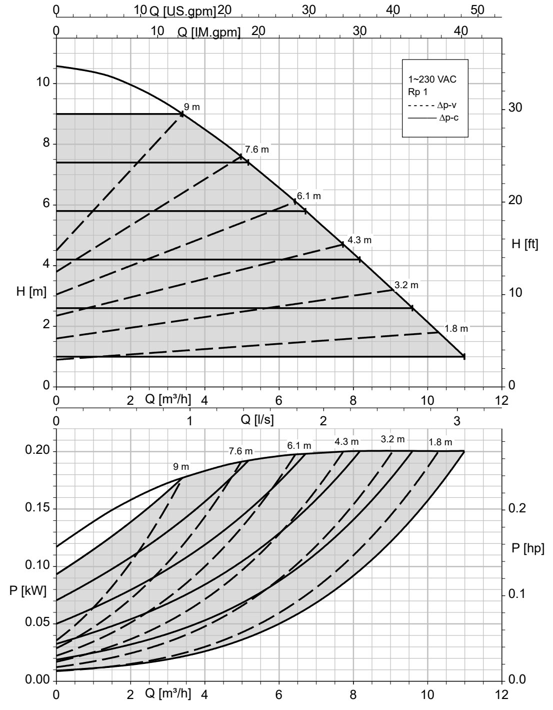
Calio Pro Plus 25-80 Δpv, Δpc



Calio Pro Plus 25-100 Open-loop Control, Dynamic Control



Calio Pro Plus 25-100 Δpv, Δpc



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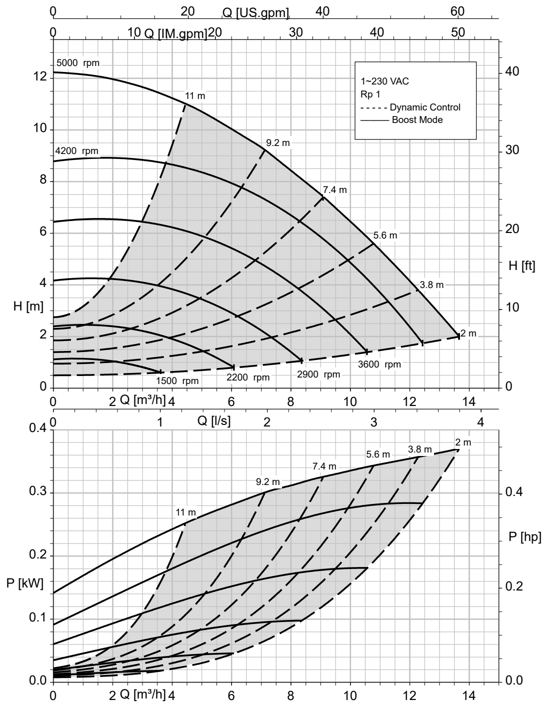
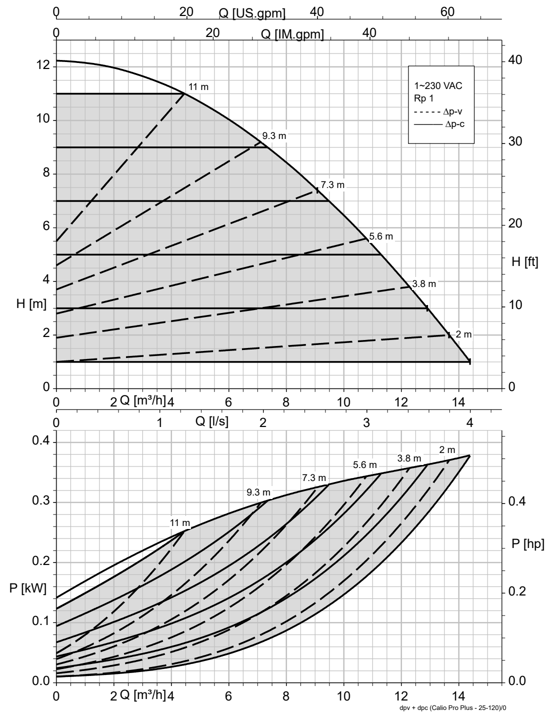
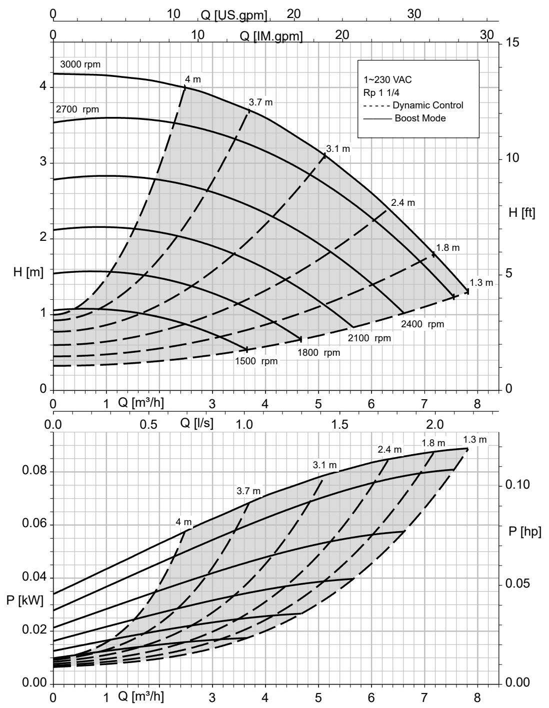
Calio Pro Plus 25-120 Open-loop Control, Dynamic Control



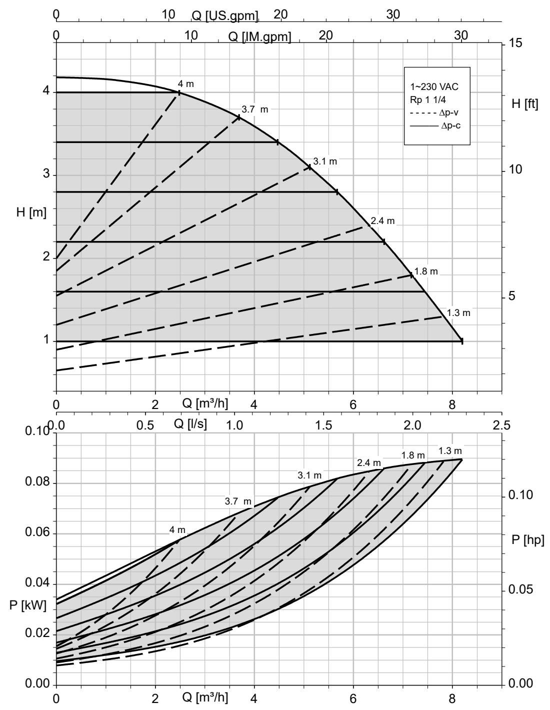
Calio Pro Plus 25-120 Δpv, Δpc



Calio Pro Plus 30-40 Open-loop Control, Dynamic Control



Calio Pro Plus 30-40 Δpv, Δpc



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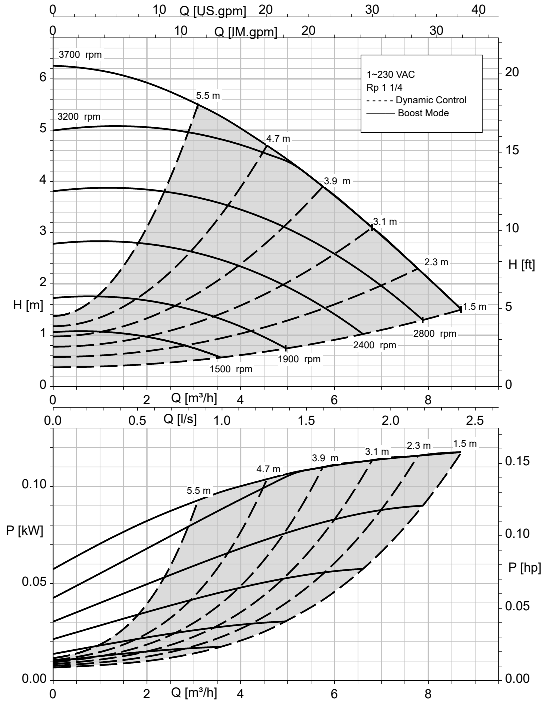
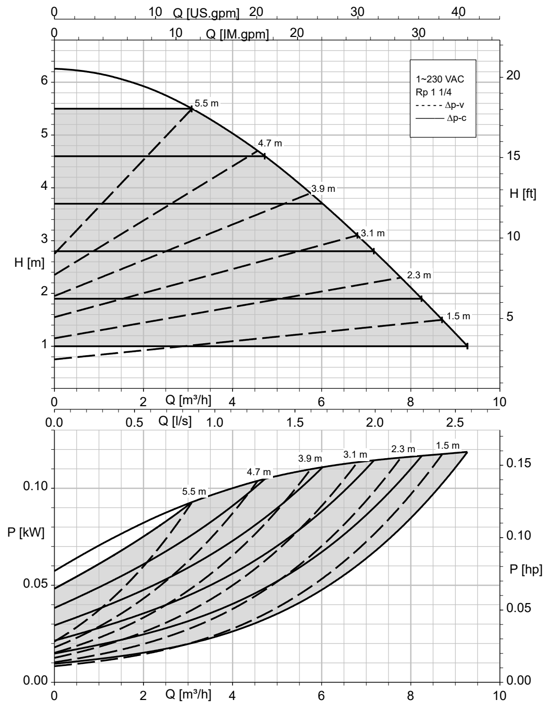
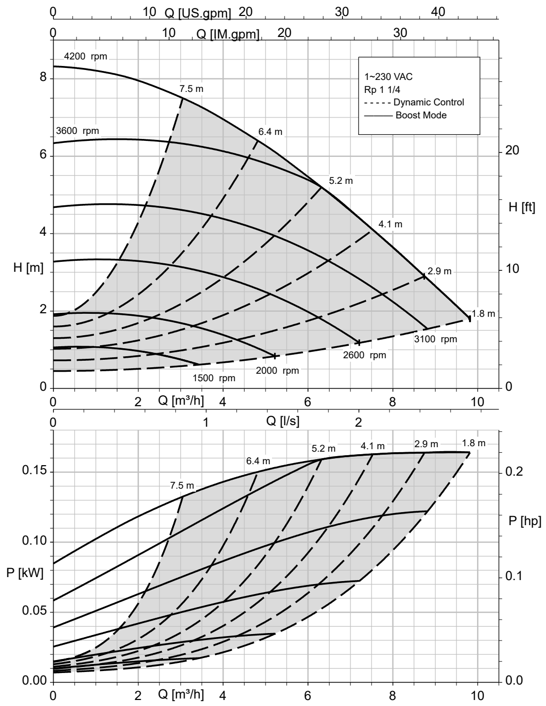
Calio Pro Plus 30-60 Open-loop Control, Dynamic Control



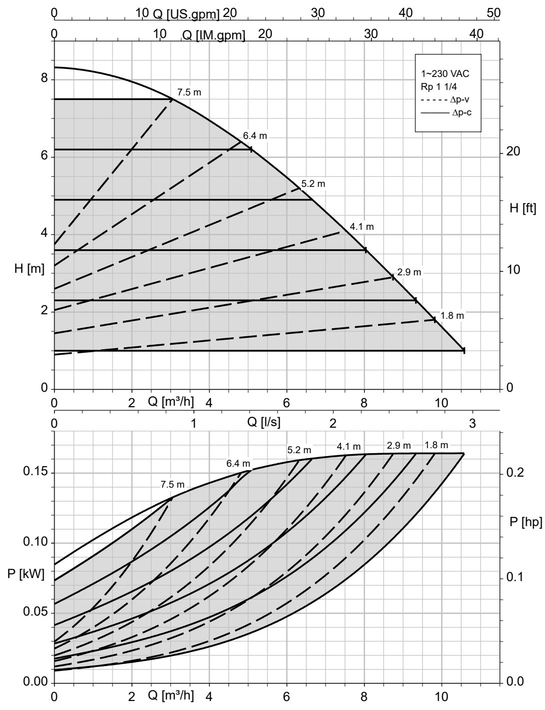
Calio Pro Plus 30-60 Δpv, Δpc



Calio Pro Plus 30-80 Open-loop Control, Dynamic Control



Calio Pro Plus 30-80 Δpv, Δpc



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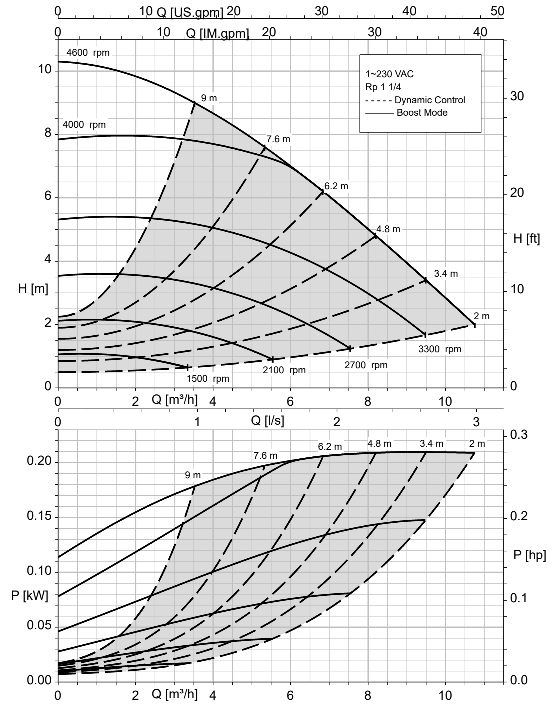
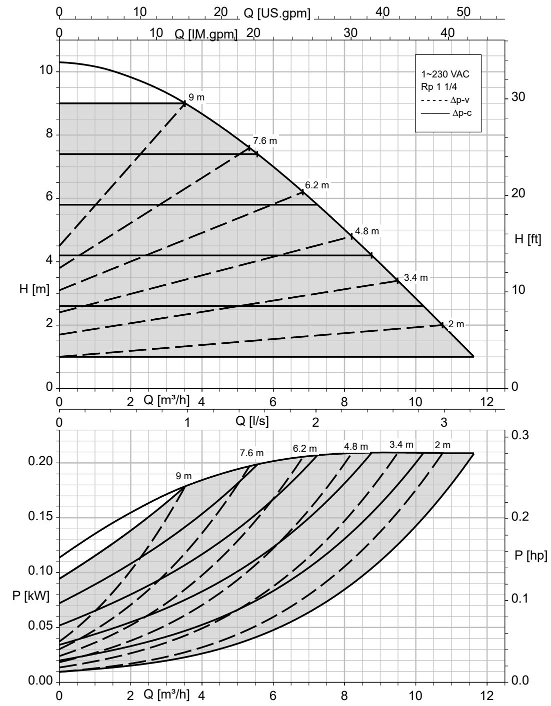
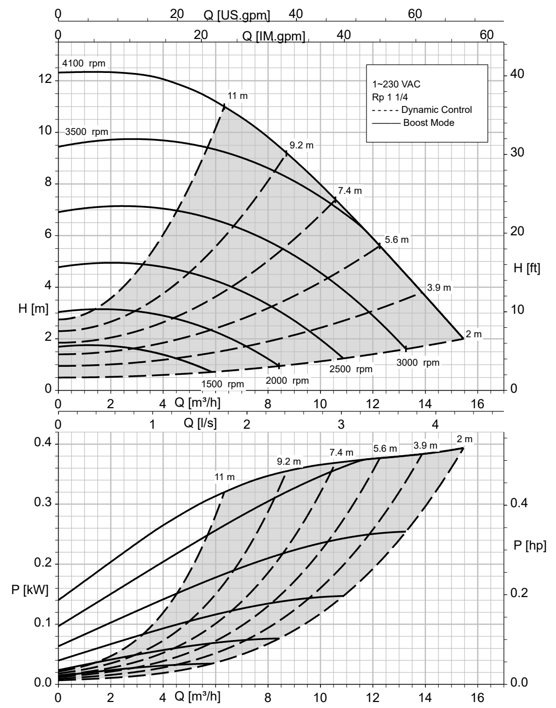
Calio Pro Plus 30-100 Open-loop Control, Dynamic Control



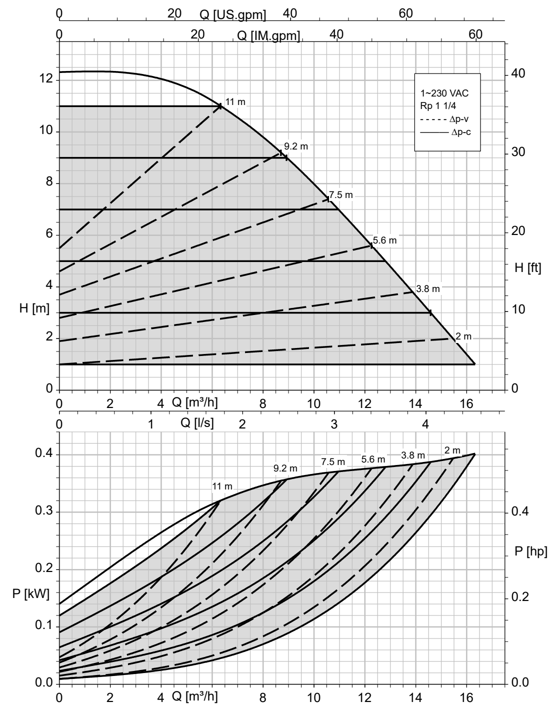
Calio Pro Plus 30-100 Δpv, Δpc



Calio Pro Plus 30-120 Open-loop Control, Dynamic Control



Calio Pro Plus 30-120 Δpv, Δpc



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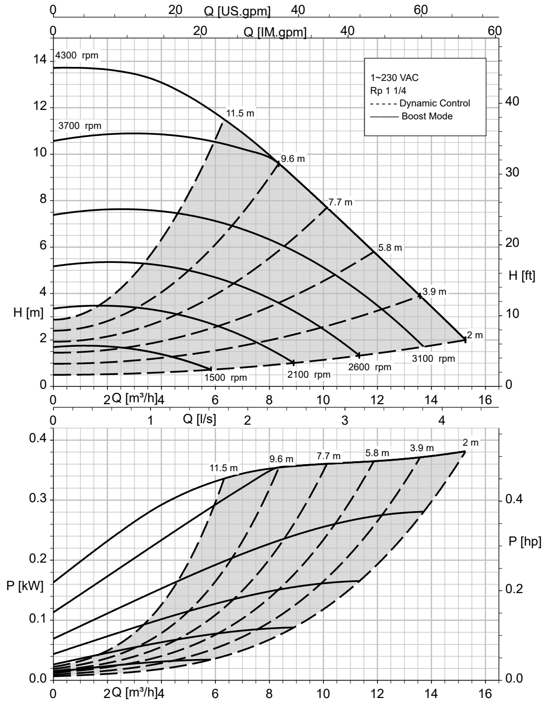
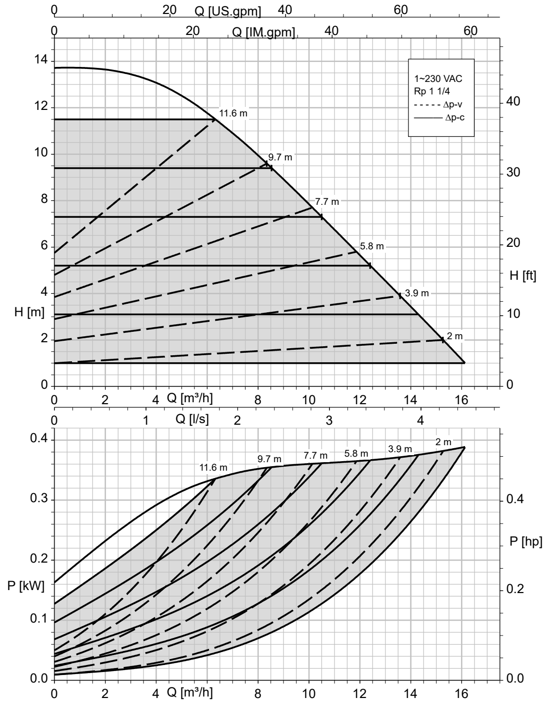
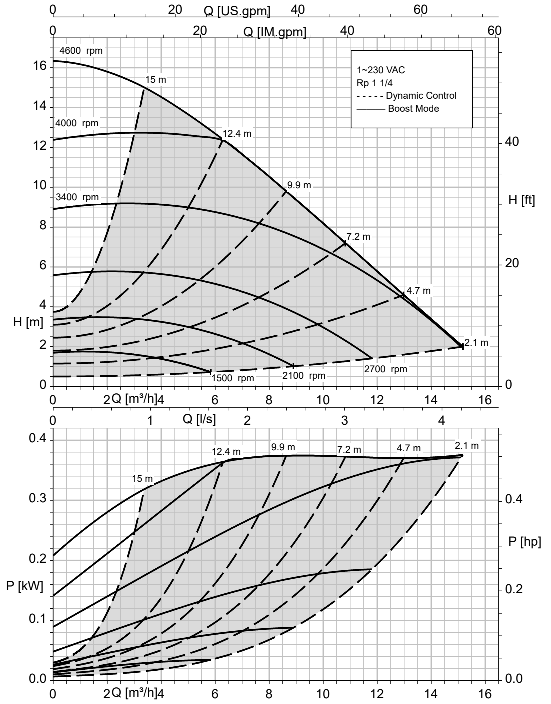
Calio Pro Plus 30-140 Open-loop Control, Dynamic Control



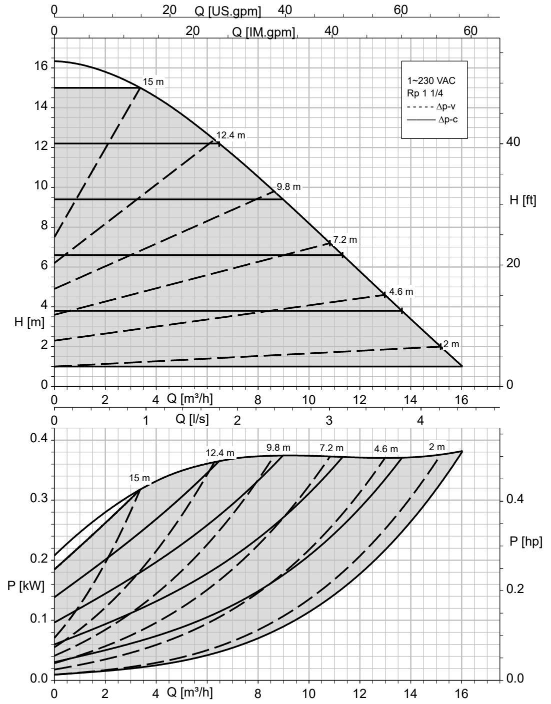
Calio Pro Plus 30-140 Δpv, Δpc



Calio Pro Plus 30-160 Open-loop Control, Dynamic Control



Calio Pro Plus 30-160 Δpv, Δpc



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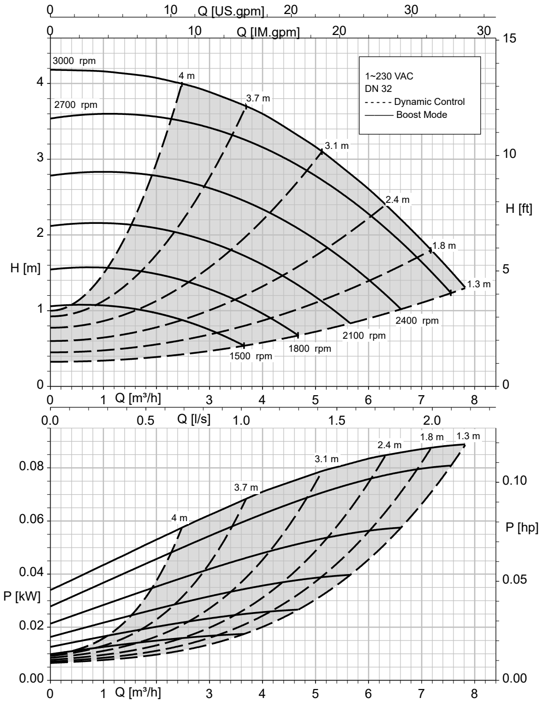
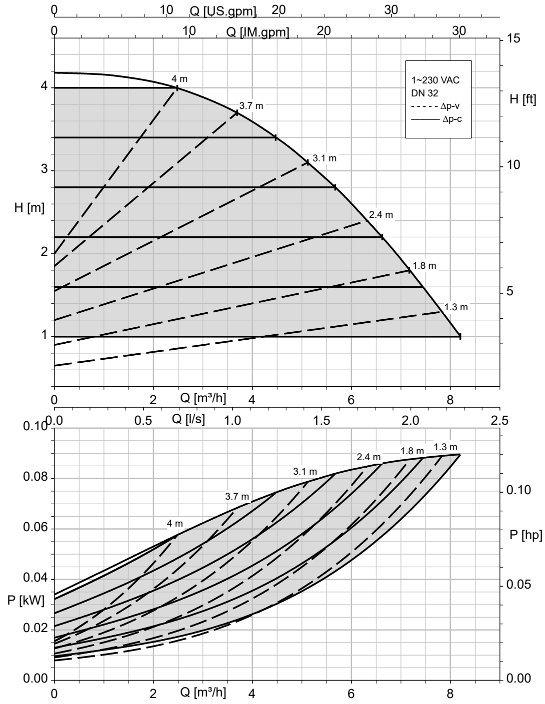
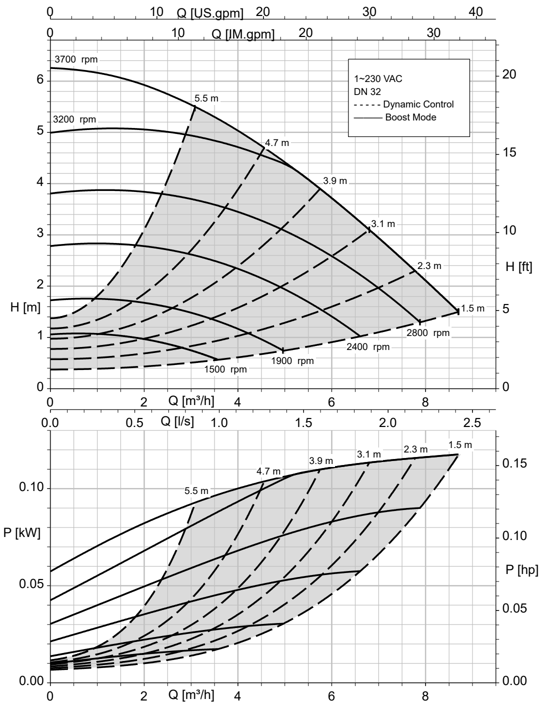
Calio Pro Plus 32-40 Open-loop Control, Dynamic Control



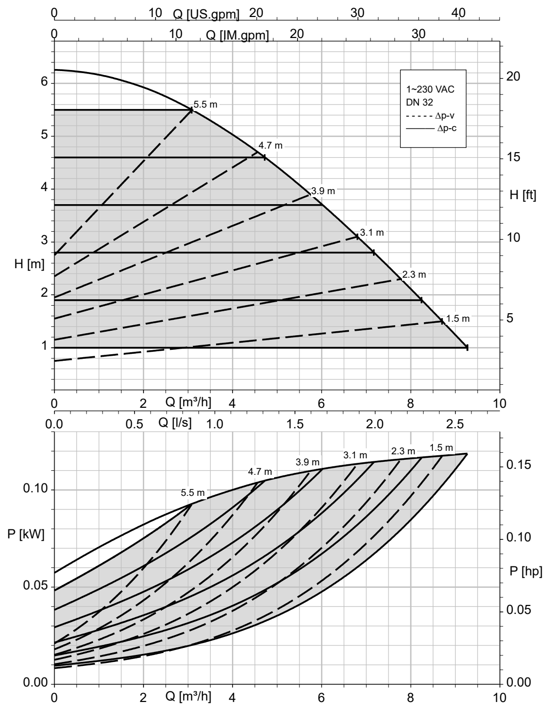
Calio Pro Plus 32-40 Δpv, Δpc



Calio Pro Plus 32-60 Open-loop Control, Dynamic Control



Calio Pro Plus 32-60 Δpv, Δpc



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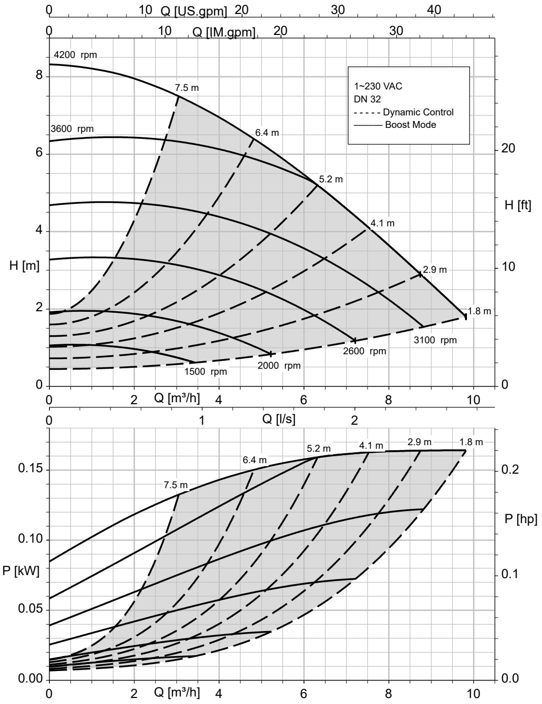
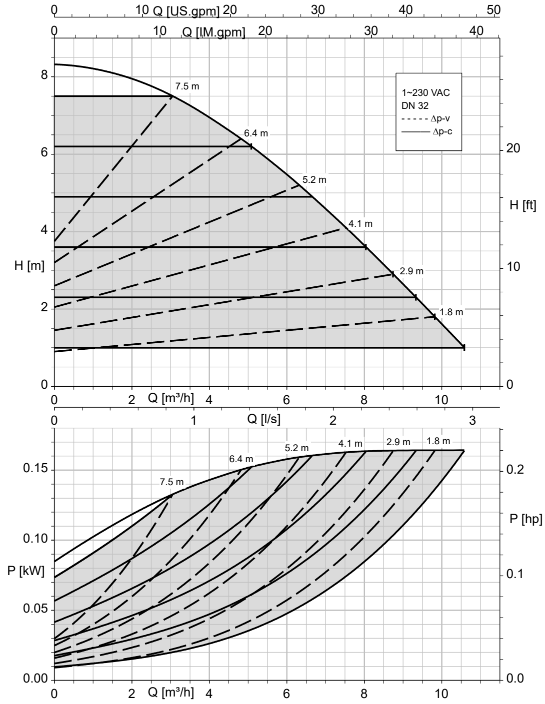
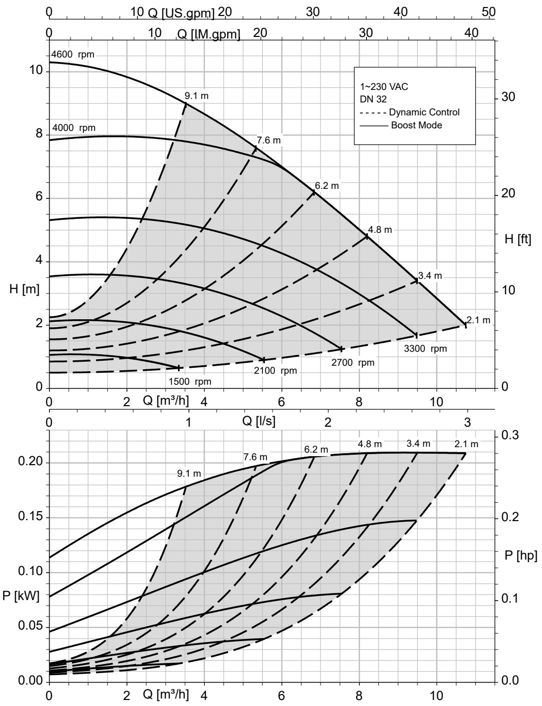
Calio Pro Plus 32-80 Open-loop Control, Dynamic Control



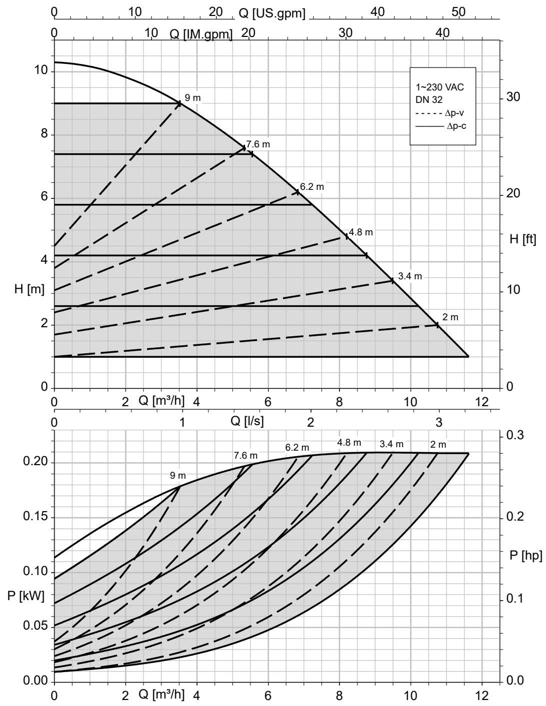
Calio Pro Plus 32-80 Δpv, Δpc



Calio Pro Plus 32-100 Open-loop Control, Dynamic Control



Calio Pro Plus 32-100 Δpv, Δpc



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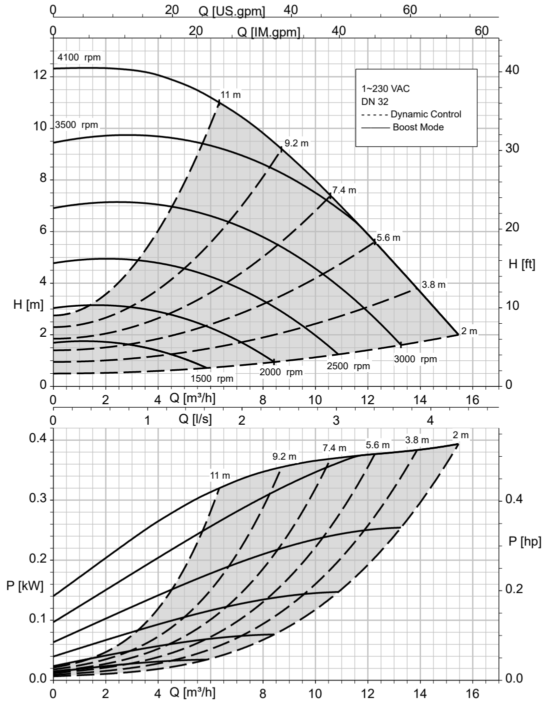
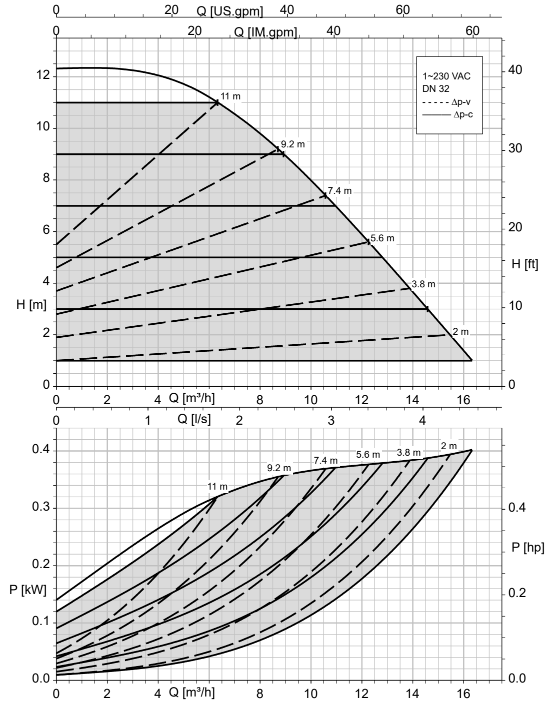
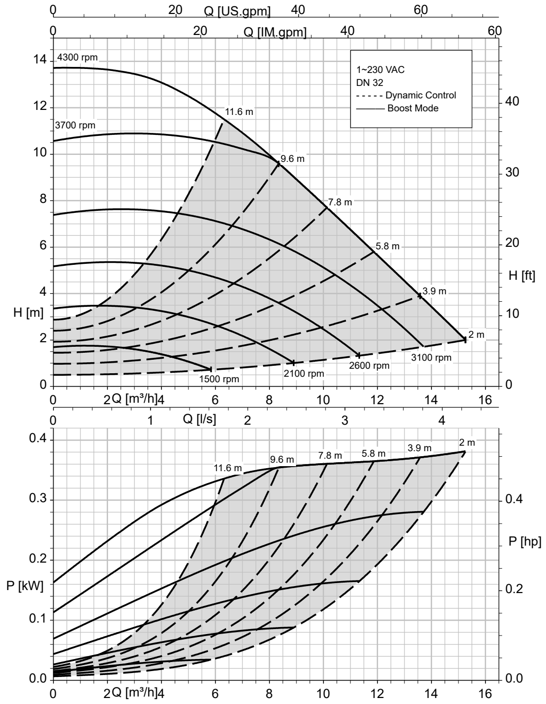
Calio Pro Plus 32-120 Open-loop Control, Dynamic Control



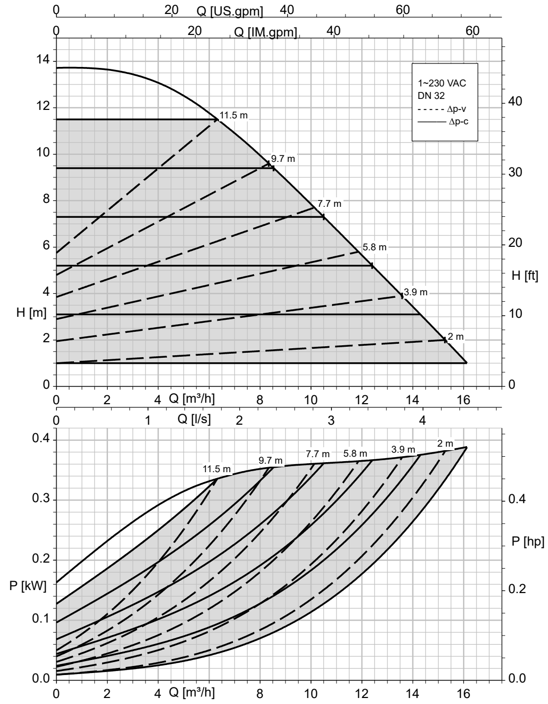
Calio Pro Plus 32-120 Δpv, Δpc



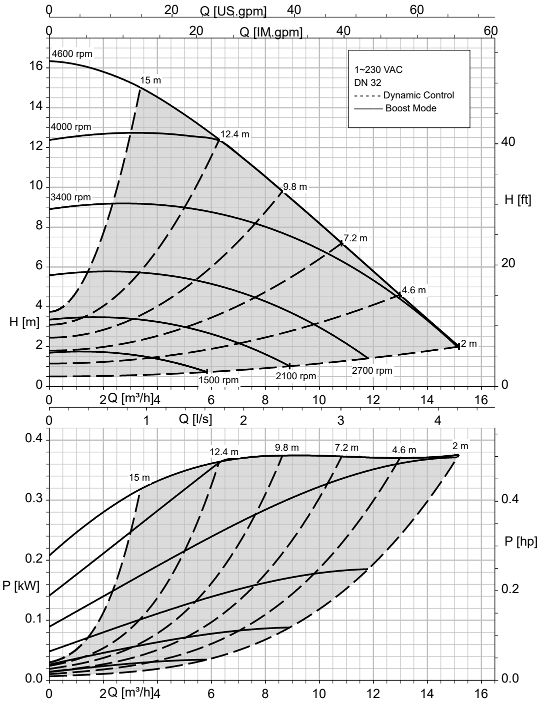
Calio Pro Plus 32-140 Open-loop Control, Dynamic Control



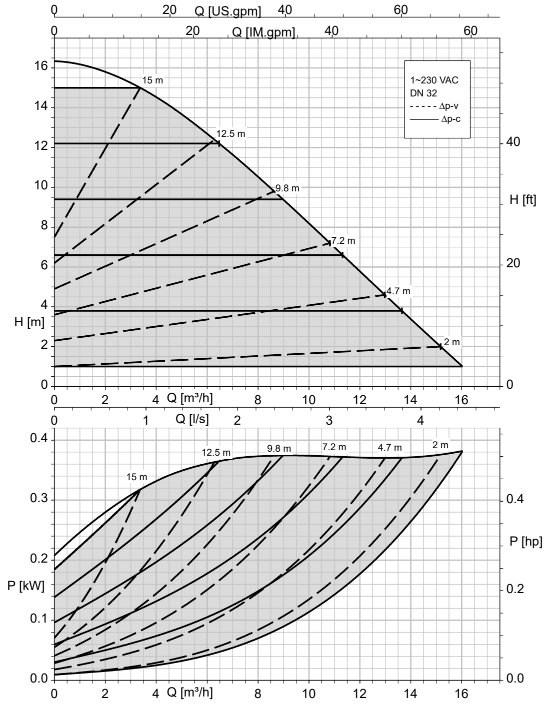
Calio Pro Plus 32-140 Δpv, Δpc



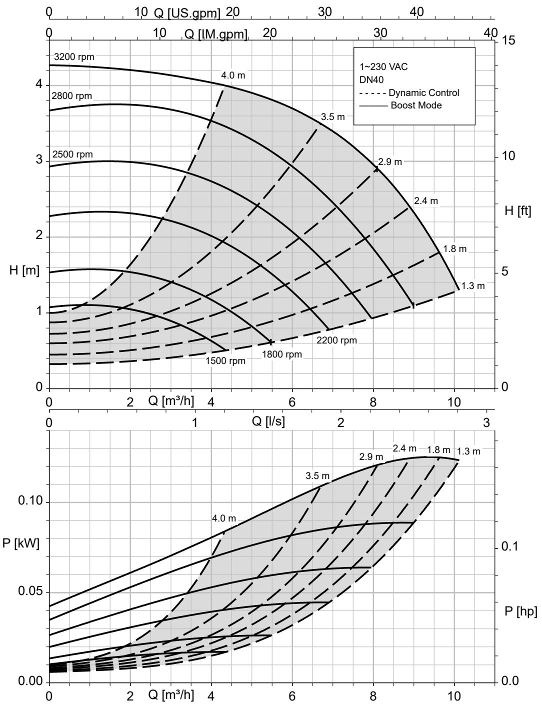
Calio Pro Plus 32-160 Open-loop Control, Dynamic Control



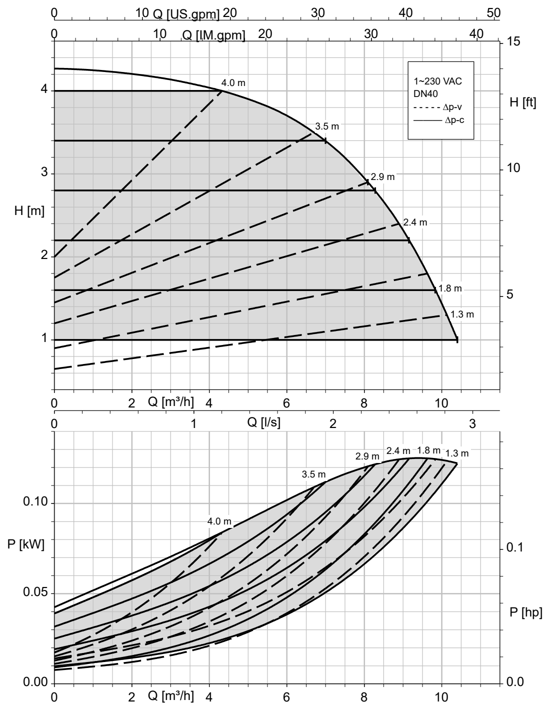
Calio Pro Plus 32-160 Δpv, Δpc



Calio Pro Plus 40-40 Open-loop Control, Dynamic Control



Calio Pro Plus 40-40 Δpv, Δpc



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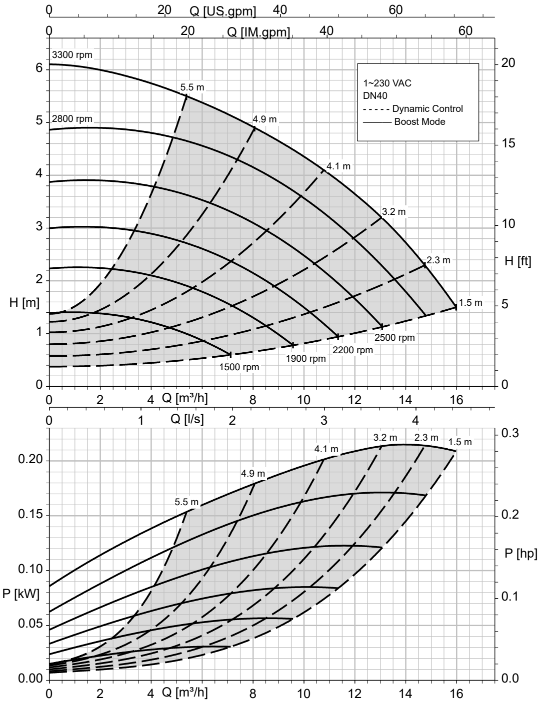
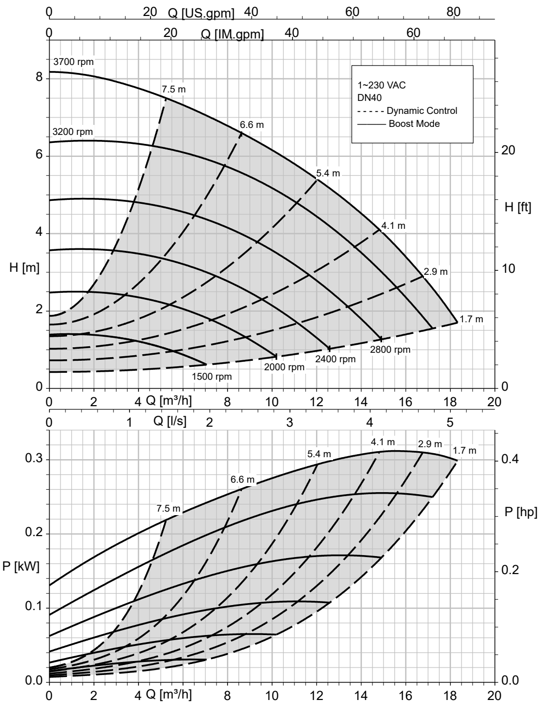
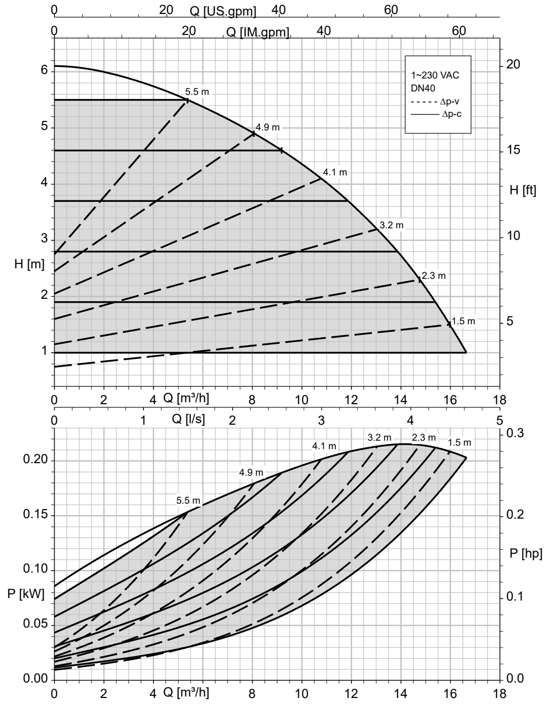
Calio Pro Plus 40-60 Open-loop Control, Dynamic Control



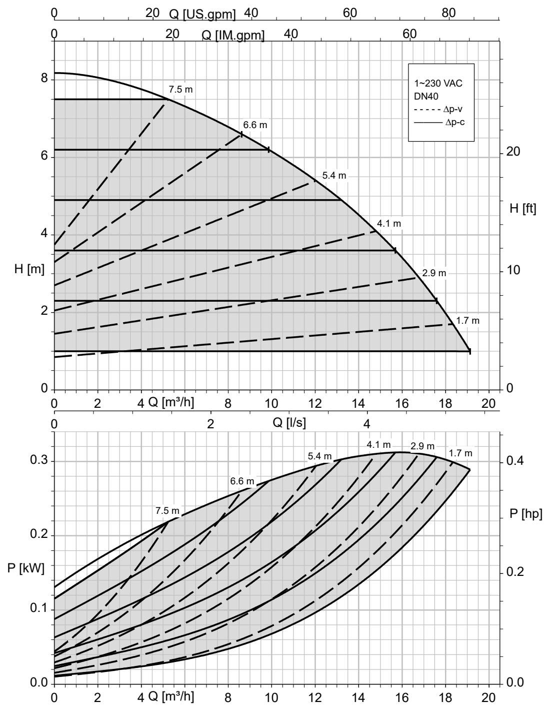
Calio Pro Plus 40-80 Open-loop Control, Dynamic Control



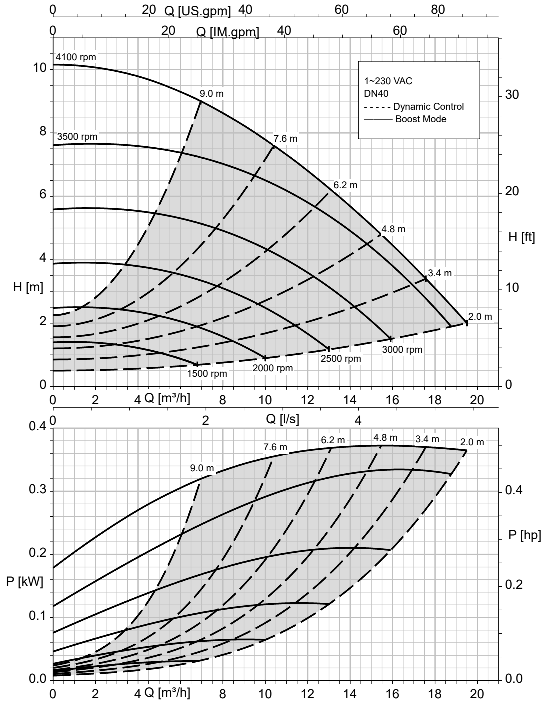
Calio Pro Plus 40-60 Δpv, Δpc



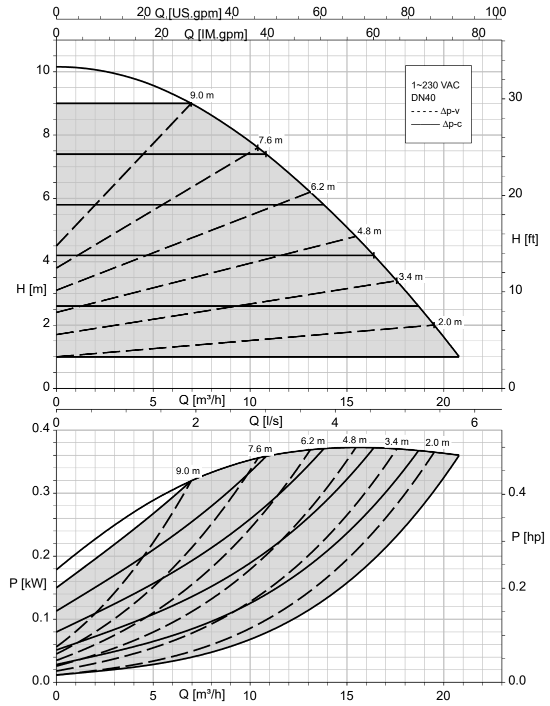
Calio Pro Plus 40-80 Δpv, Δpc



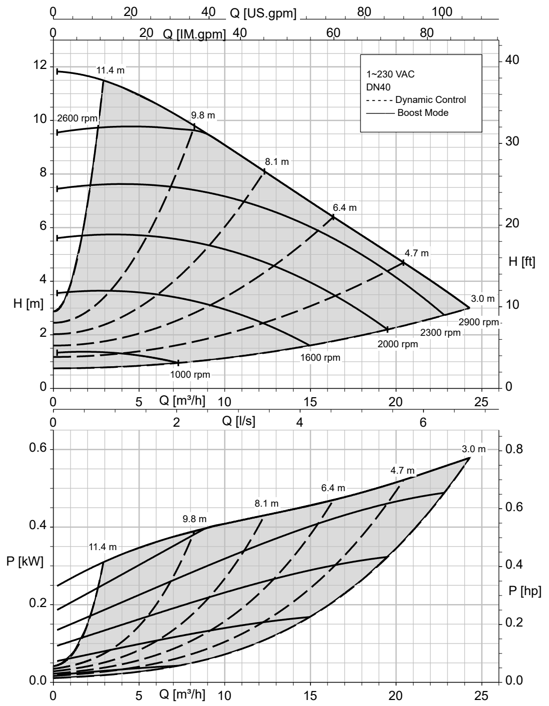
Calio Pro Plus 40-100 Open-loop Control, Dynamic Control



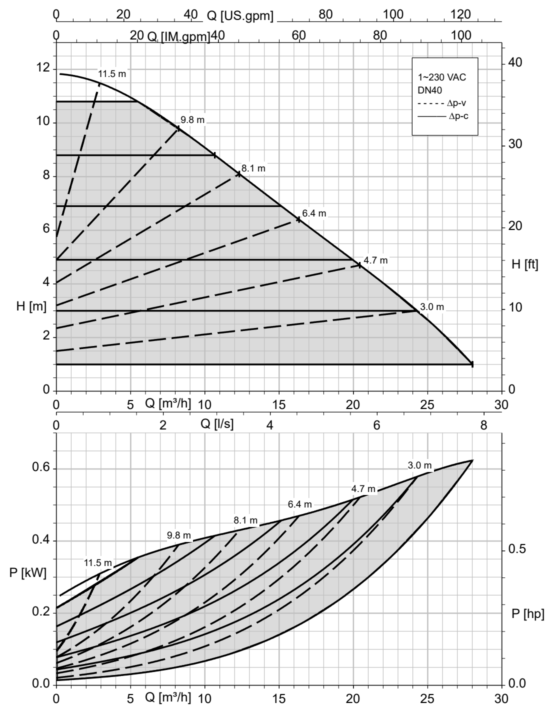
Calio Pro Plus 40-100 Δpv, Δpc



Calio Pro Plus 40-120 Open-loop Control, Dynamic Control



Calio Pro Plus 40-120 Δpv, Δpc



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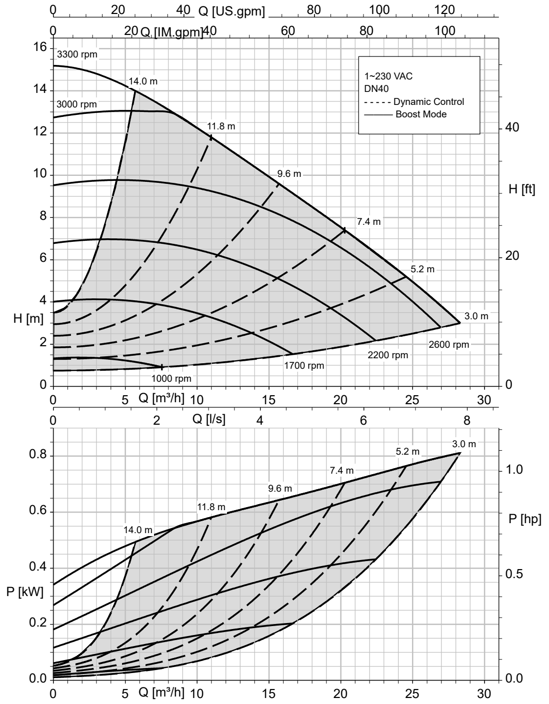
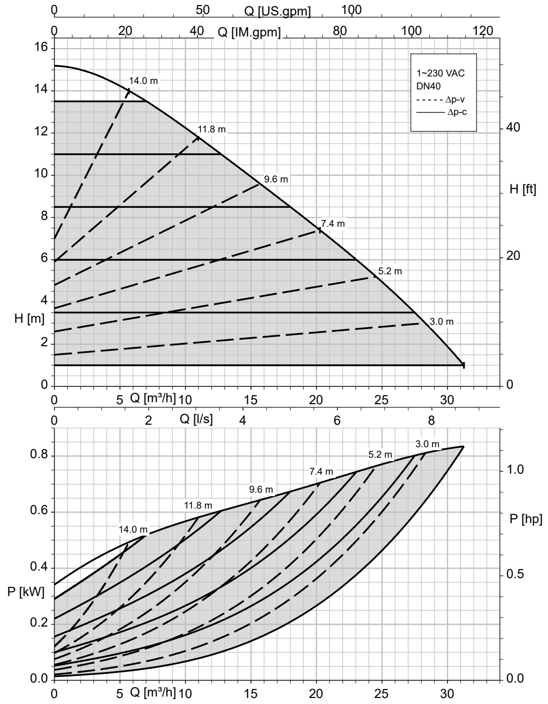
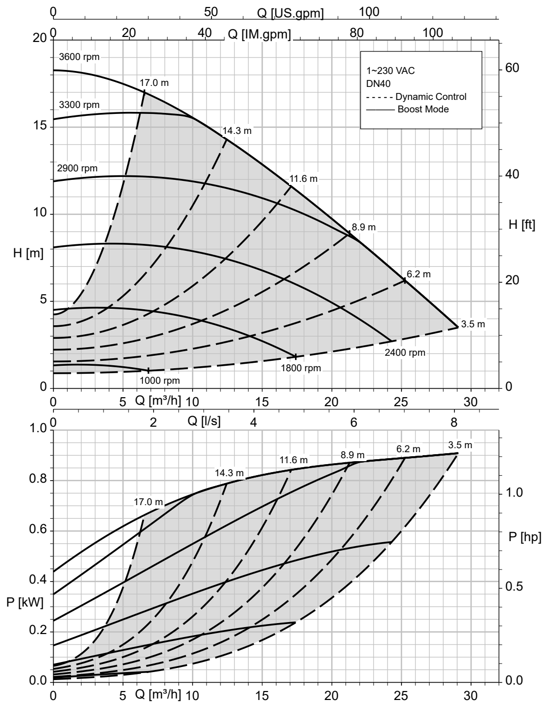
Calio Pro Plus 40-150 Open-loop Control, Dynamic Control



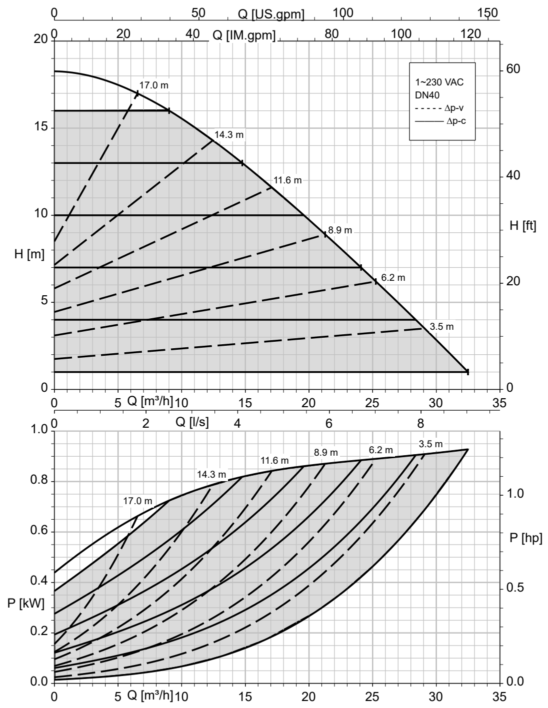
Calio Pro Plus 40-150 Δpv, Δpc



Calio Pro Plus 40-180 Open-loop Control, Dynamic Control



Calio Pro Plus 40-180 Δpv, Δpc



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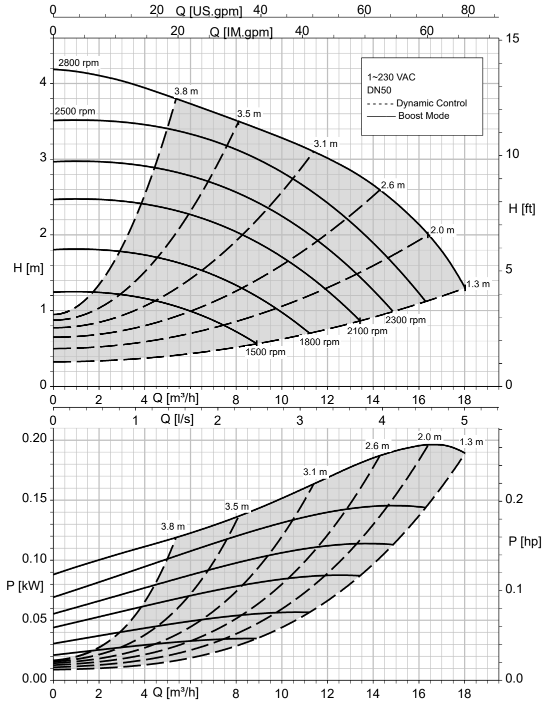
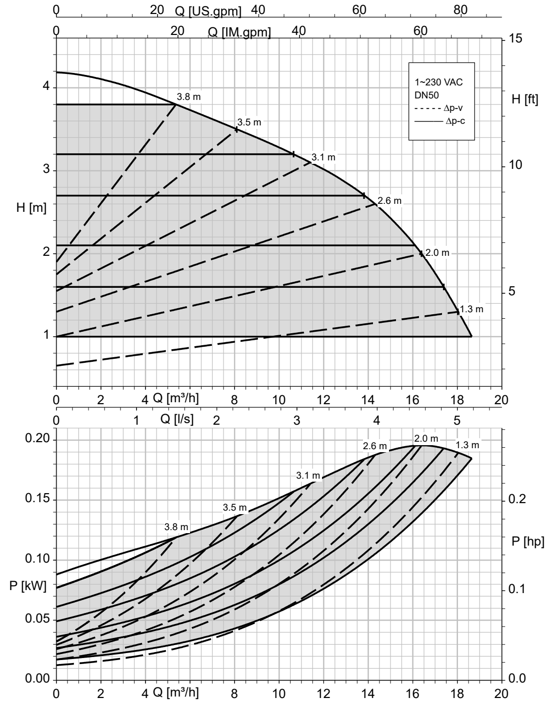
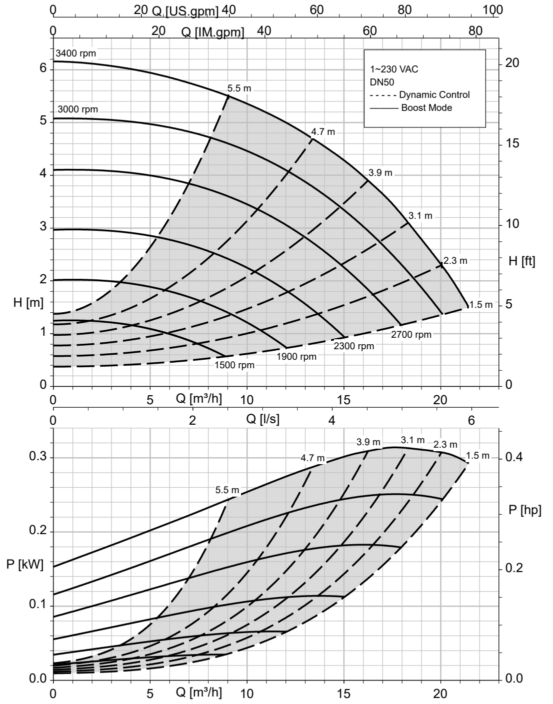
Calio Pro Plus 50-40 Open-loop Control, Dynamic Control



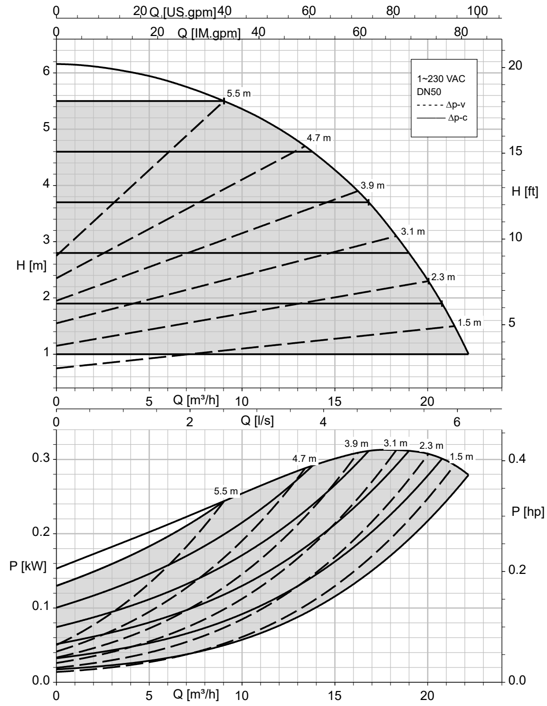
Calio Pro Plus 50-40 Δpv, Δpc



Calio Pro Plus 50-60 Open-loop control, Dynamic Control



Calio Pro Plus 50-60 Δpv, Δpc



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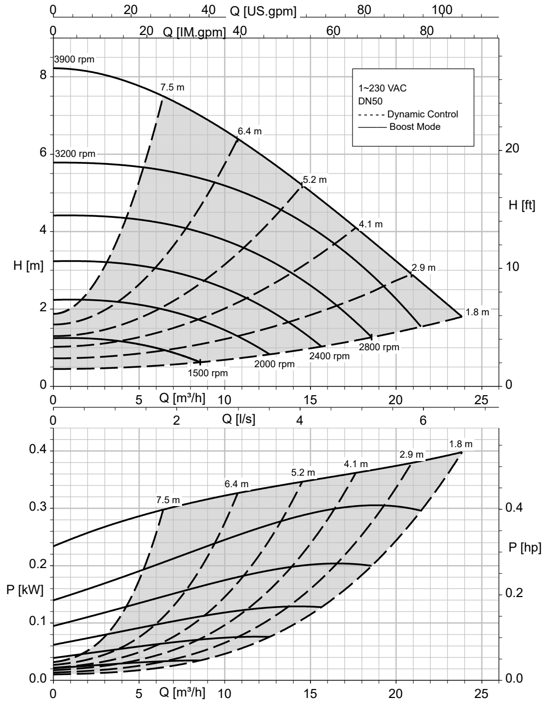
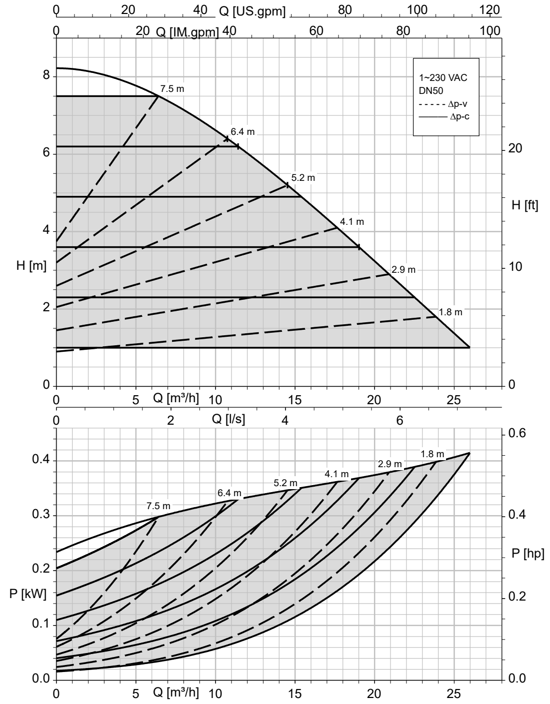
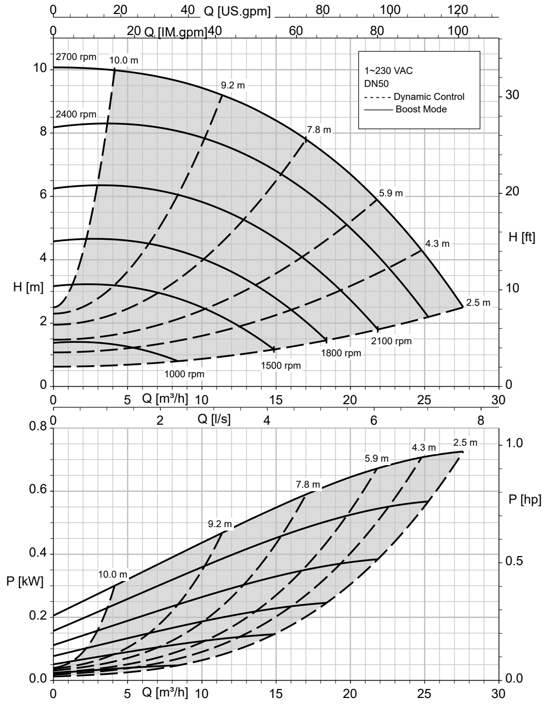
Calio Pro Plus 50-80 Open-loop control, Dynamic Control



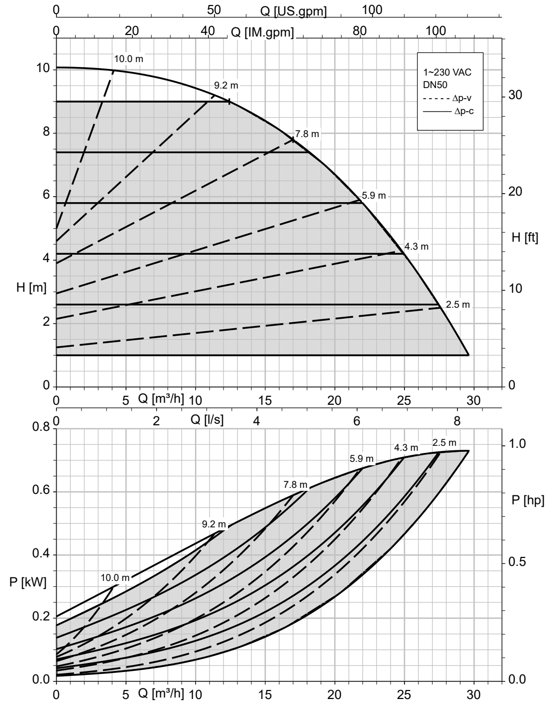
Calio Pro Plus 50-80 Δpv, Δpc



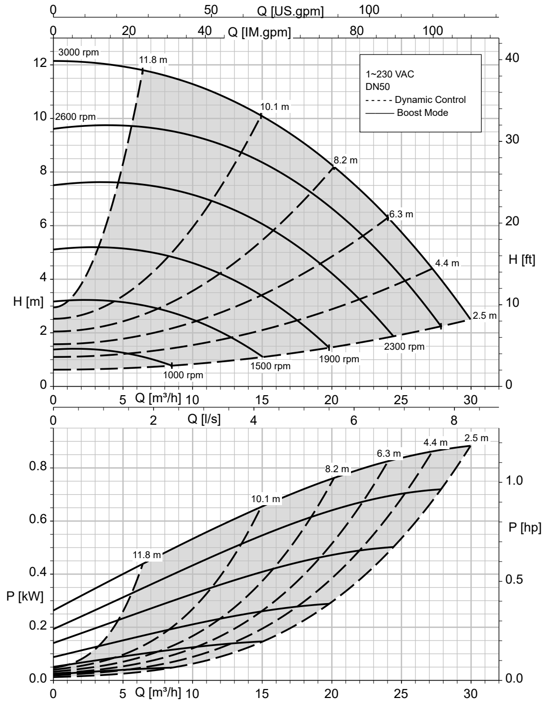
Calio Pro Plus 50-100 Open-loop Control, Dynamic Control



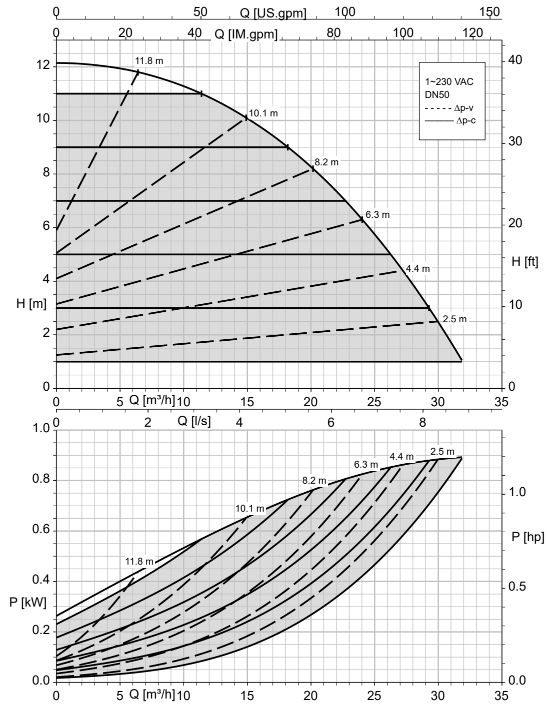
Calio Pro Plus 50-100 Δpv, Δpc



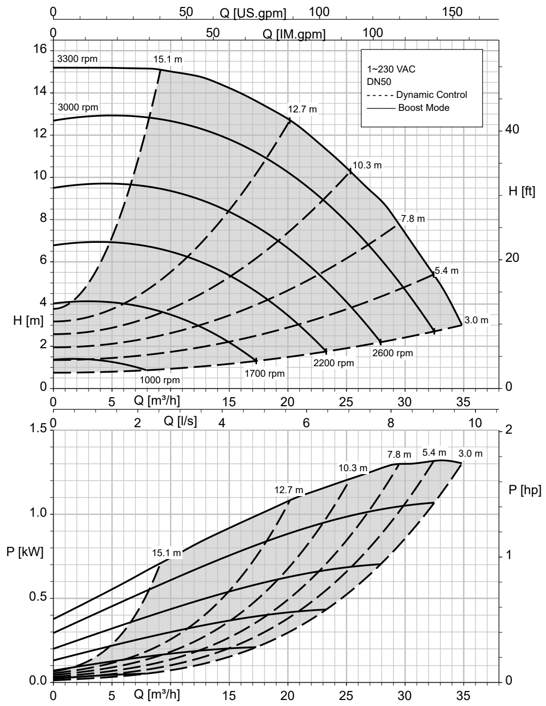
Calio Pro Plus 50-120 Open-loop Control, Dynamic Control



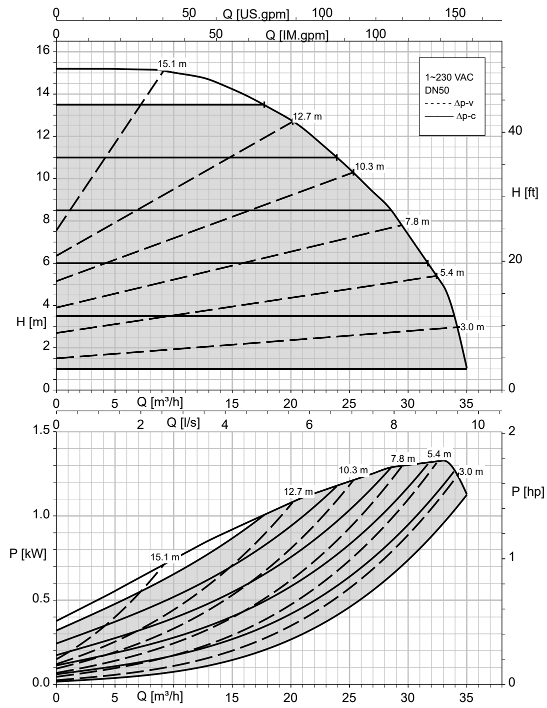
Calio Pro Plus 50-120 Δpv, Δpc



Calio Pro Plus 50-150 Open-loop control, Dynamic Control



Calio Pro Plus 50-150 Δpv, Δpc



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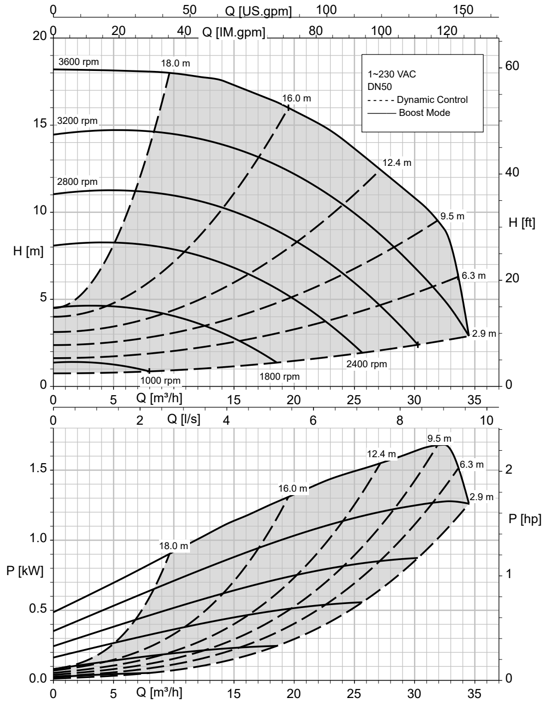
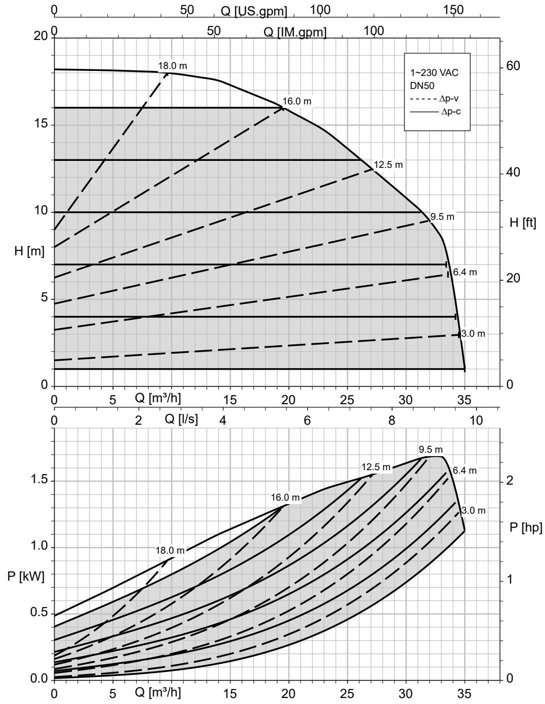
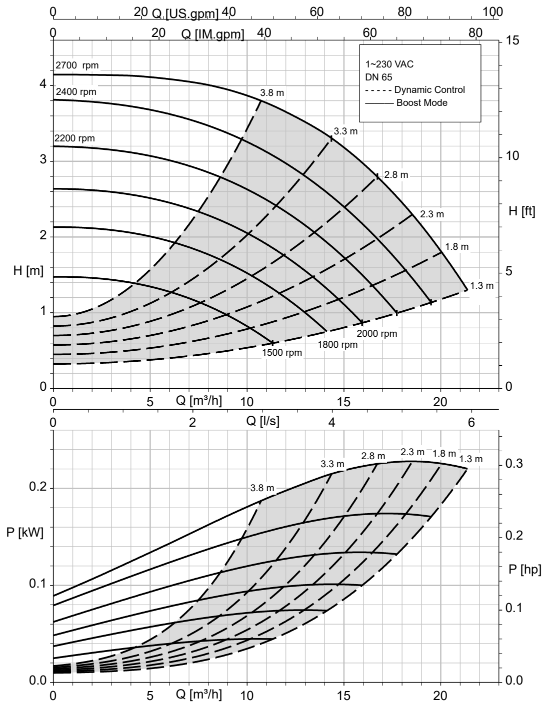
Calio Pro Plus 50-180 Open-loop control, Dynamic Control



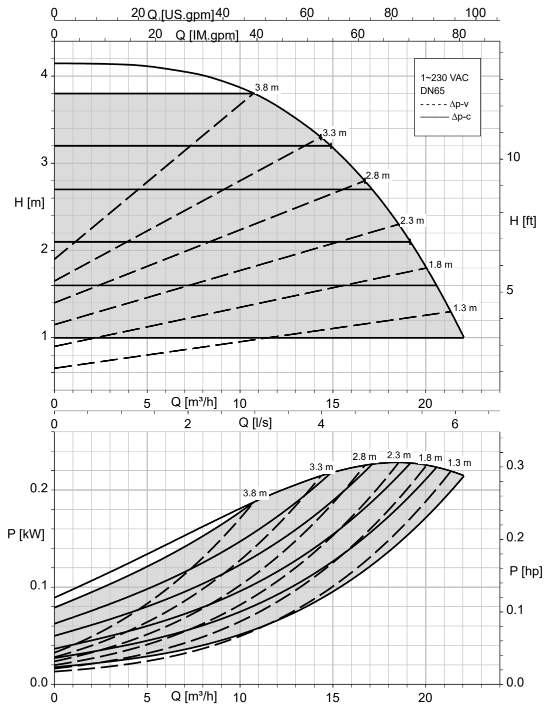
Calio Pro Plus 50-180 Δpv, Δpc



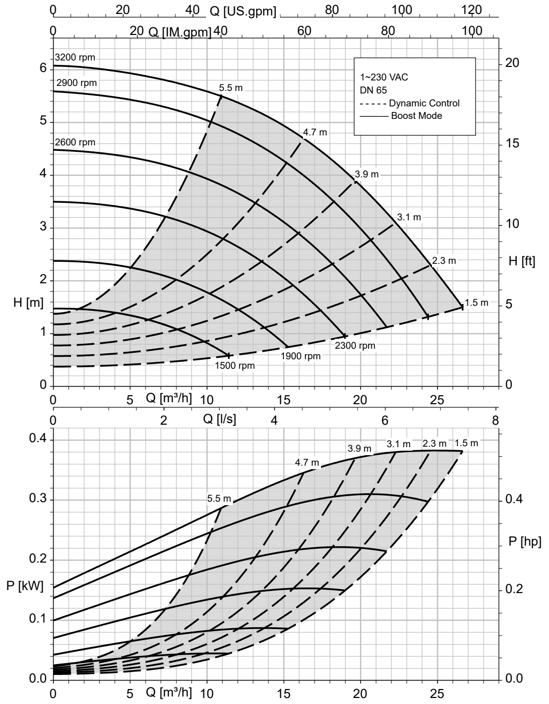
Calio Pro Plus 65-40 Open-loop Control, Dynamic Control



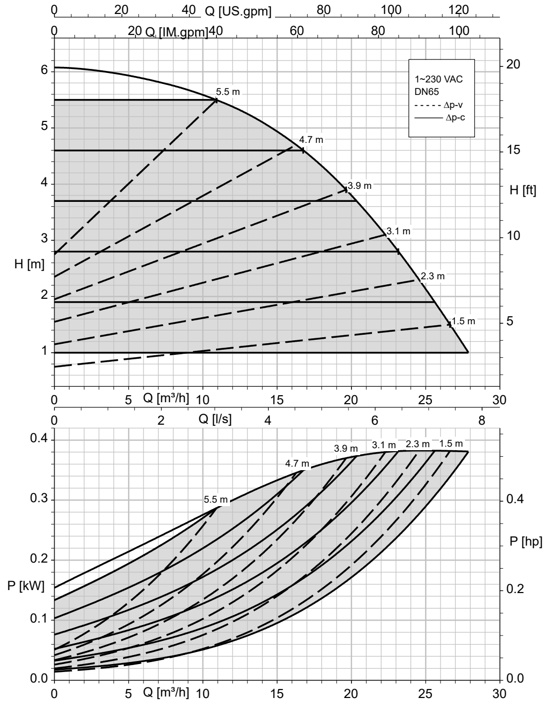
Calio Pro Plus 65-40 Δpv, Δpc



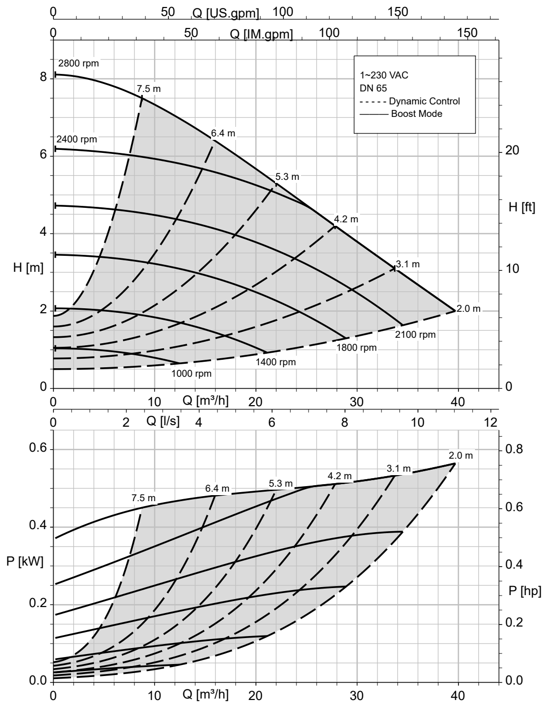
Calio Pro Plus 65-60 Open-loop Control, Dynamic Control



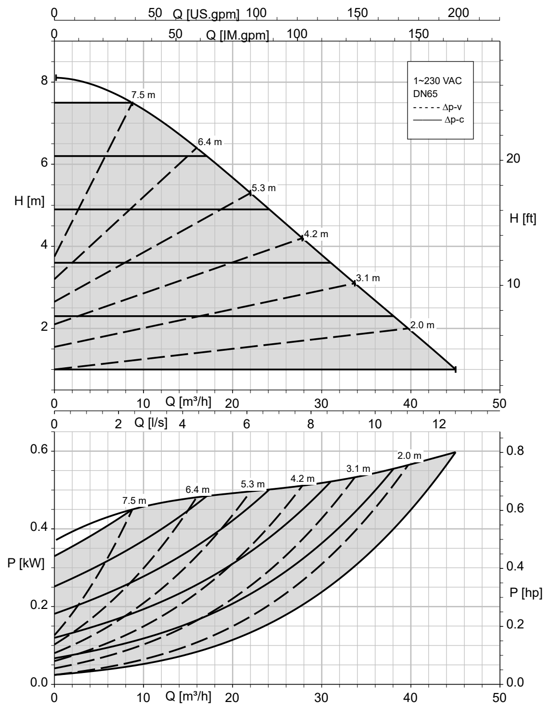
Calio Pro Plus 65-60 Δpv, Δpc



Calio Pro Plus 65-80 Open-loop Control, Dynamic Control



Calio Pro Plus 65-80 Δpv, Δpc



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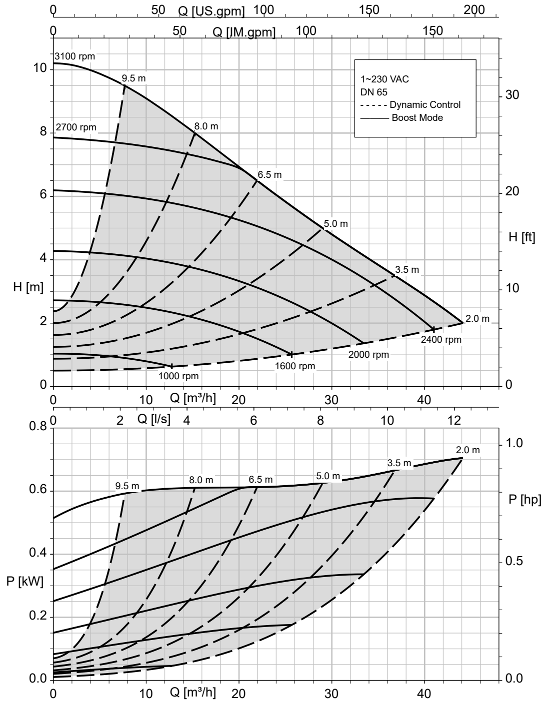
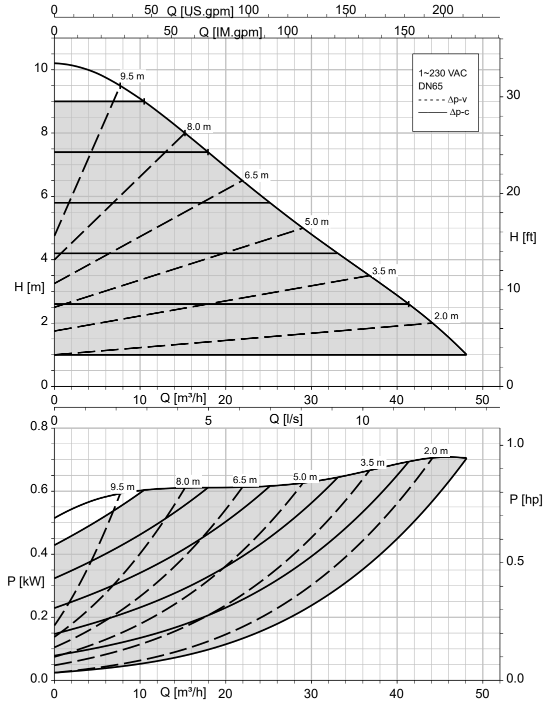
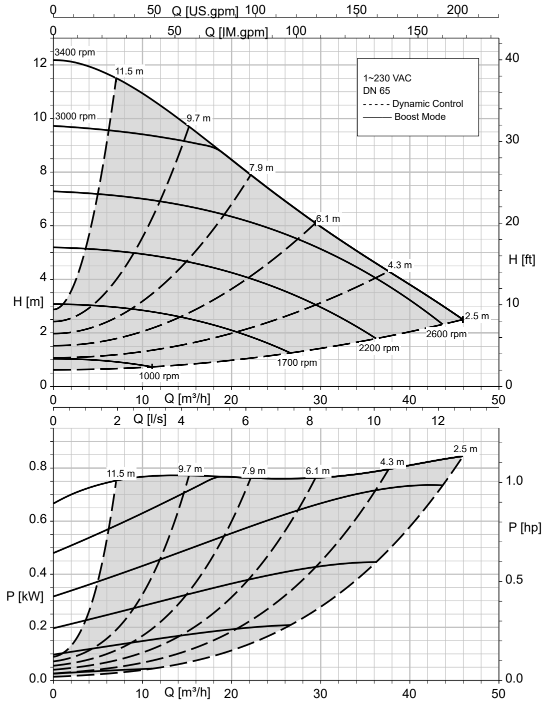
Calio Pro Plus 65-100 Open-loop Control, Dynamic Control



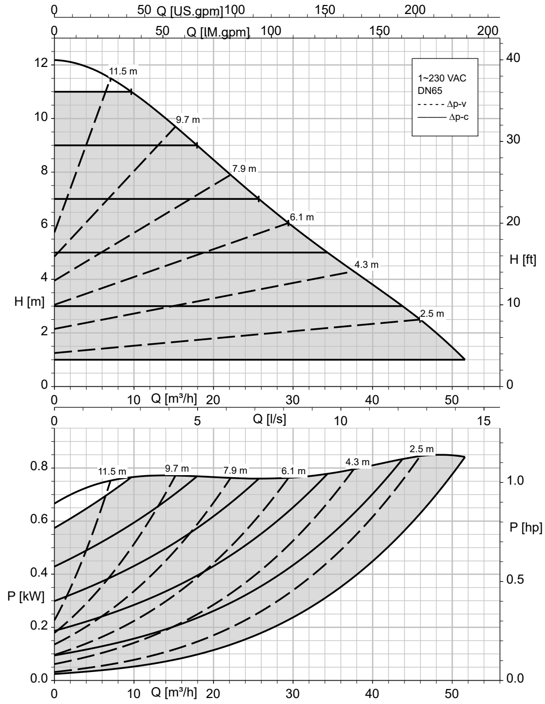
Calio Pro Plus 65-100 Δpv, Δpc



Calio Pro Plus 65-120 Open-loop Control, Dynamic Control



Calio Pro Plus 65-120 Δpv, Δpc



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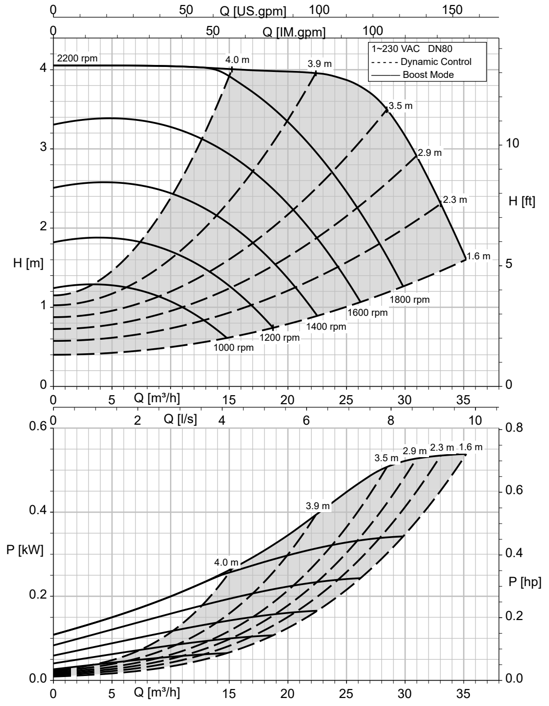
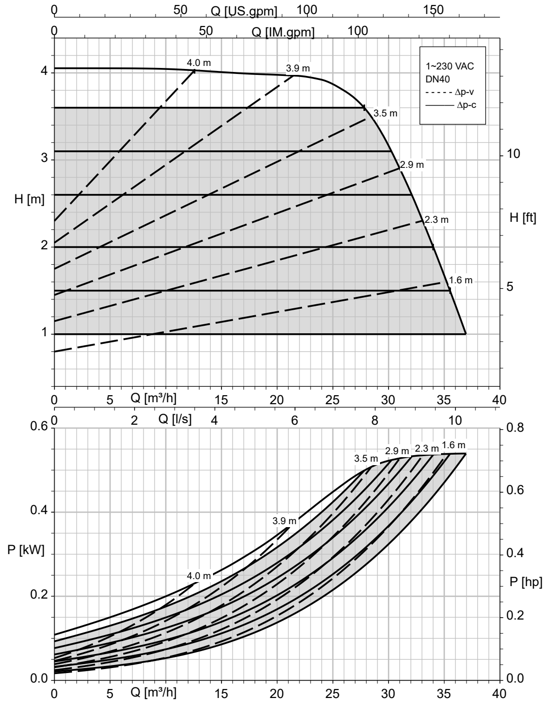
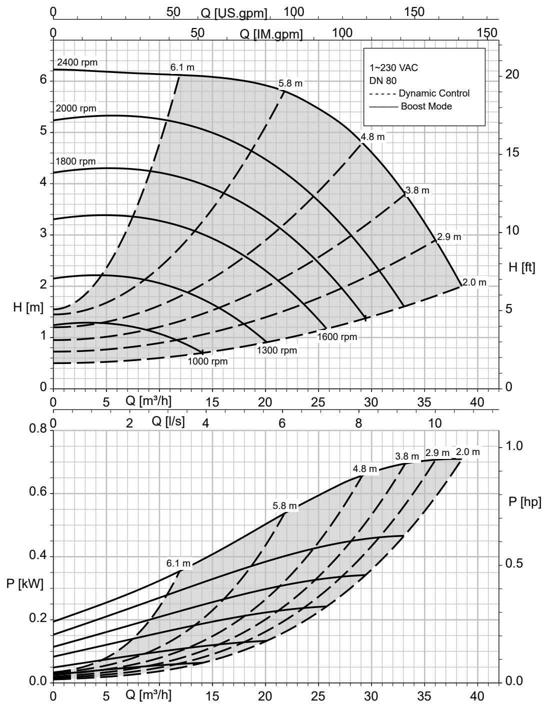
Calio Pro Plus 80-40 Open-loop Control, Dynamic Control



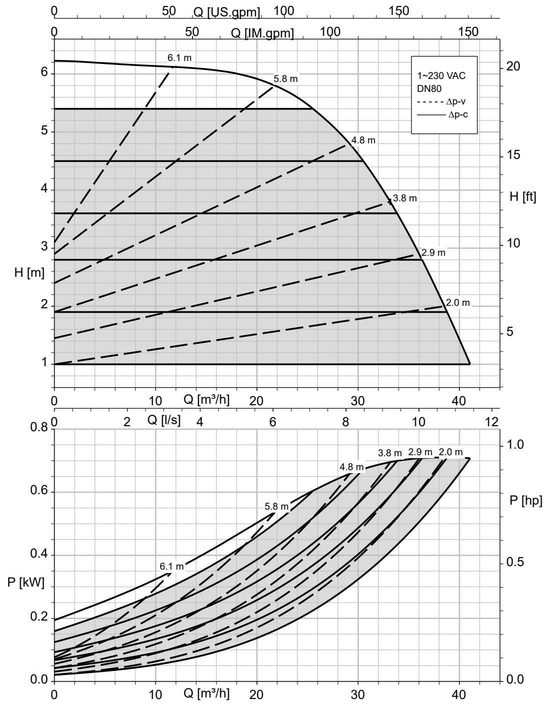
Calio Pro Plus 80-40 Δpv, Δpc



Calio Pro Plus 80-60 Open-loop Control, Dynamic Control



Calio Pro Plus 80-60 Δpv, Δpc



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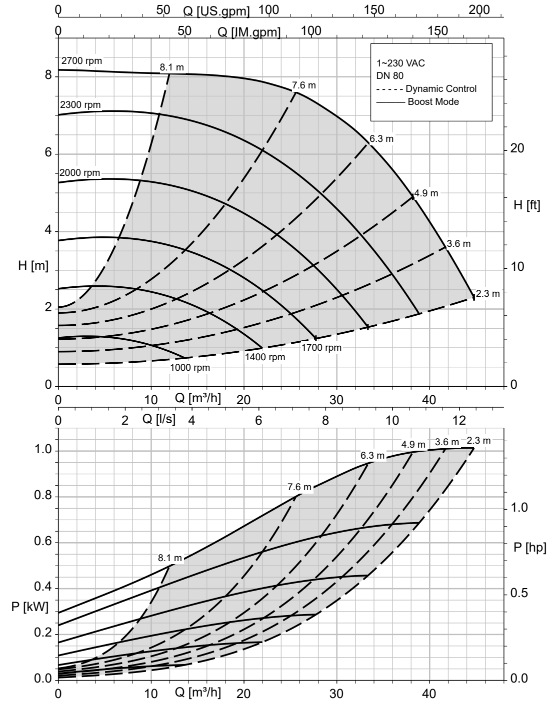
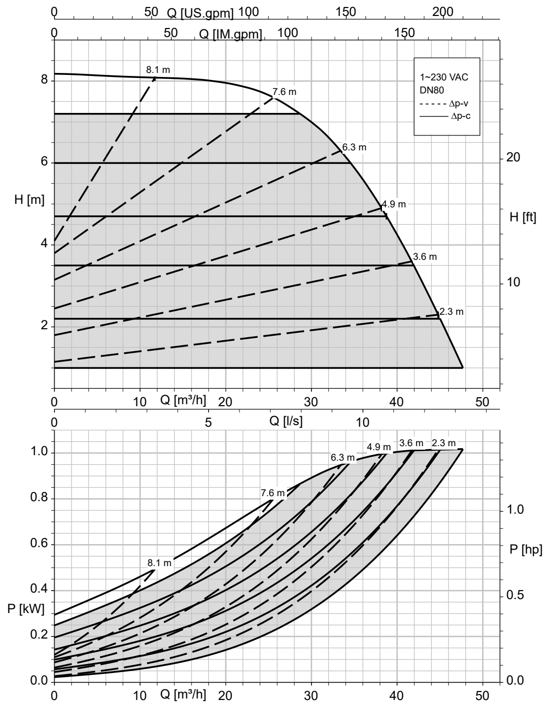
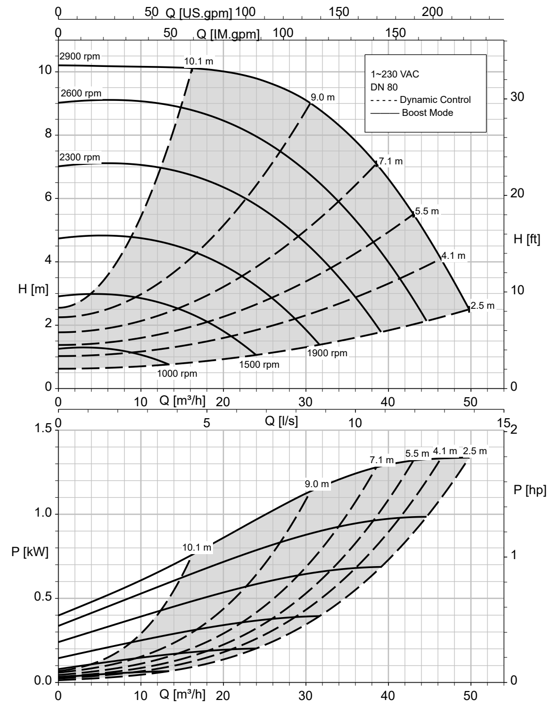
Calio Pro Plus 80-80 Open-loop control, Dynamic Control



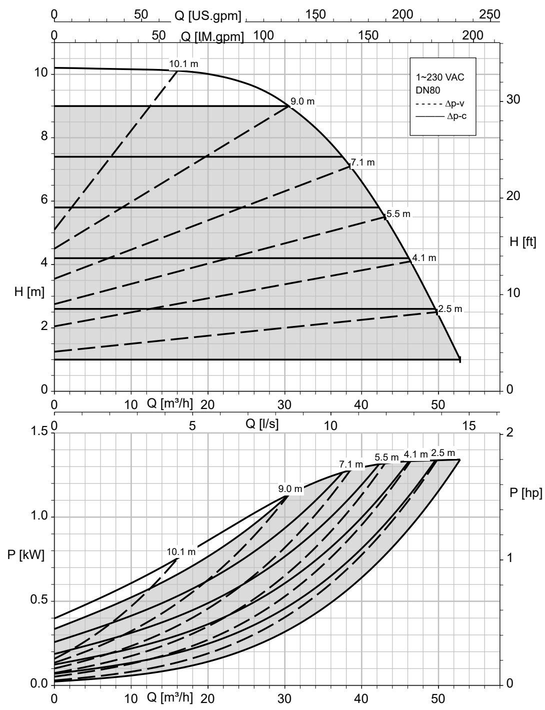
Calio Pro Plus 80-80 Δpv, Δpc



Calio Pro Plus 80-100 Open-loop Control, Dynamic Control



Calio Pro Plus 80-100 Δpv, Δpc



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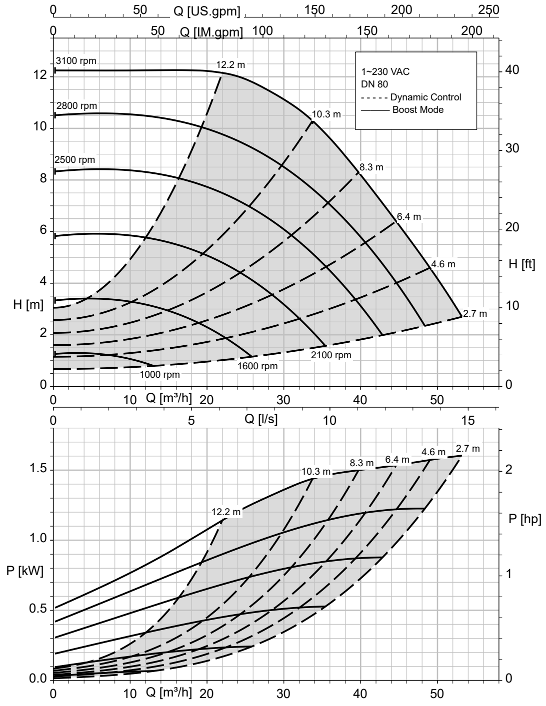
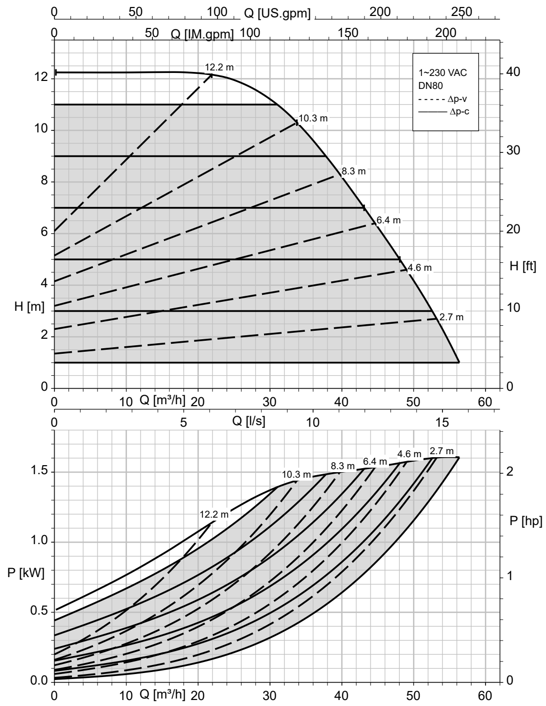
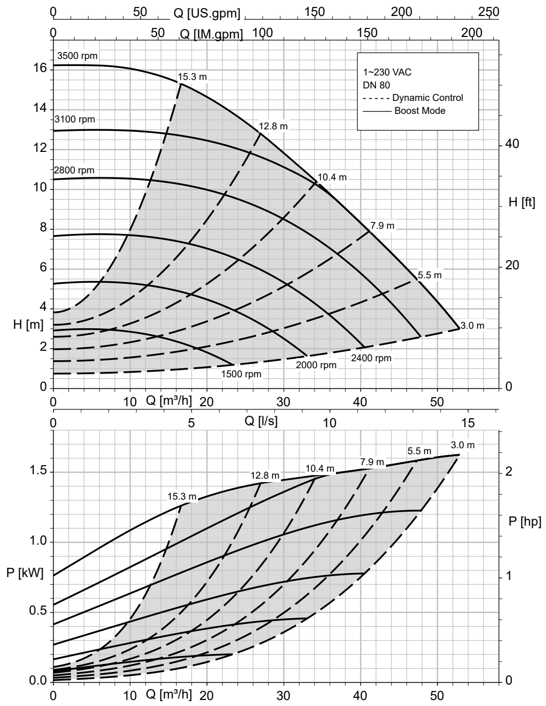
Calio Pro Plus 80-120 Open-loop Control, Dynamic Control



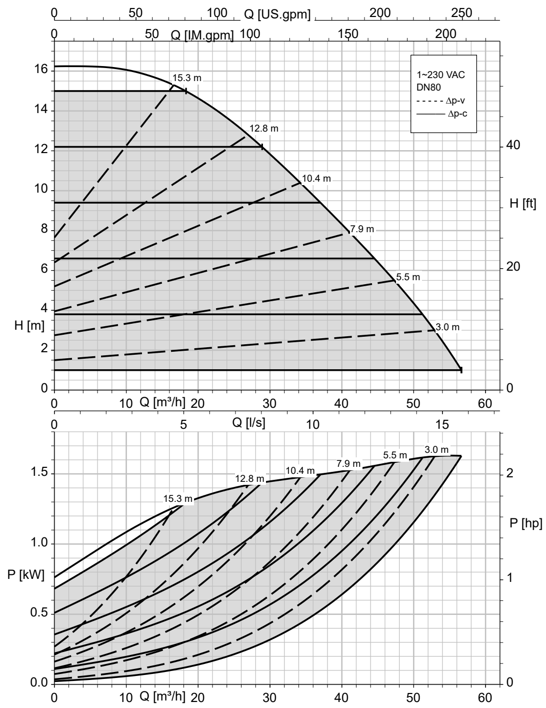
Calio Pro Plus 80-120 Δpv, Δpc



Calio Pro Plus 80-160 Open-loop Control, Dynamic Control



Calio Pro Plus 80-160 Δpv, Δpc



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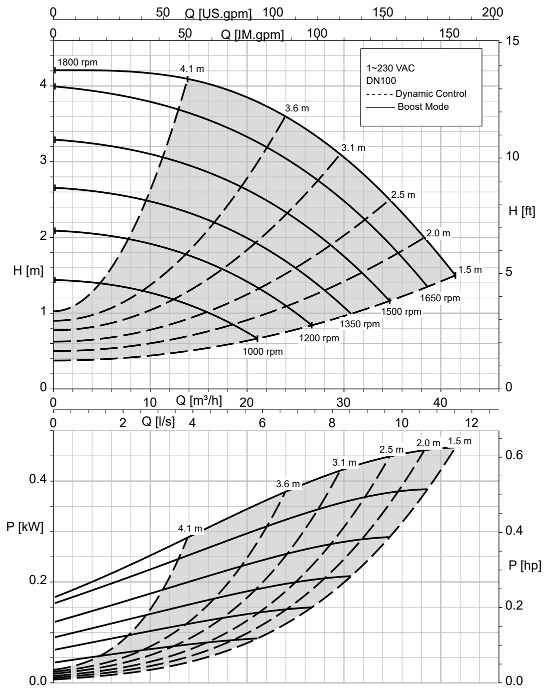
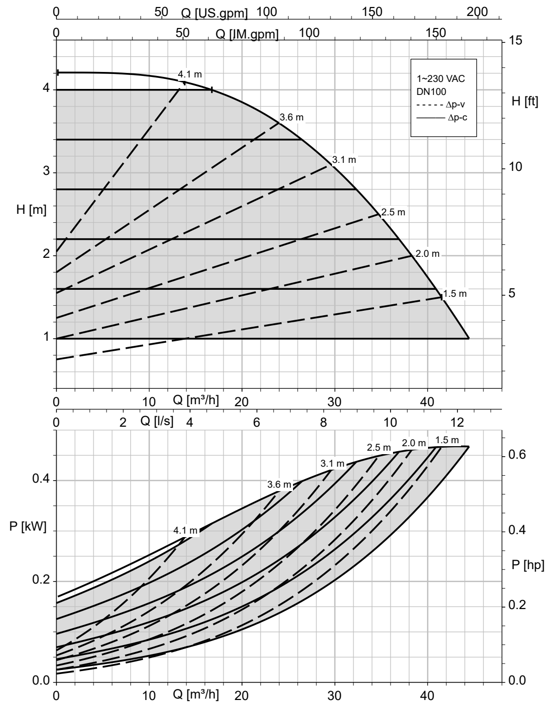
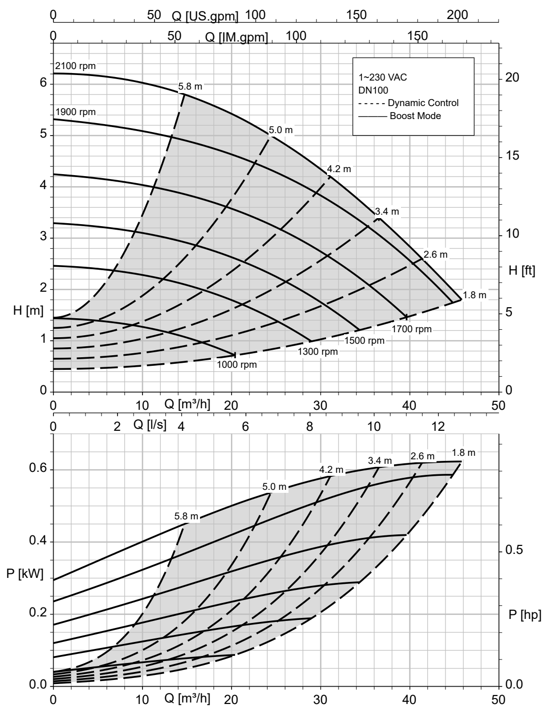
Calio Pro Plus 100-40 Open-loop Control, Dynamic Control



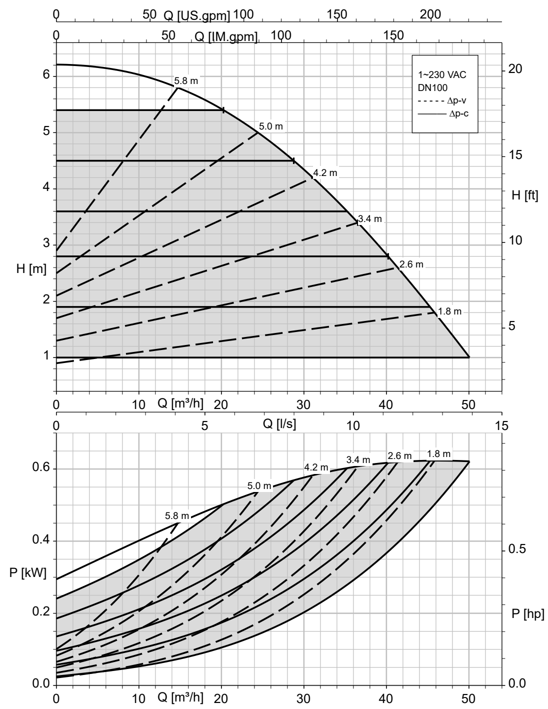
Calio Pro Plus 100-40 Δpv, Δpc



Calio Pro Plus 100-60 Open-loop control, Dynamic Control



Calio Pro Plus 100-60 Δpv, Δpc



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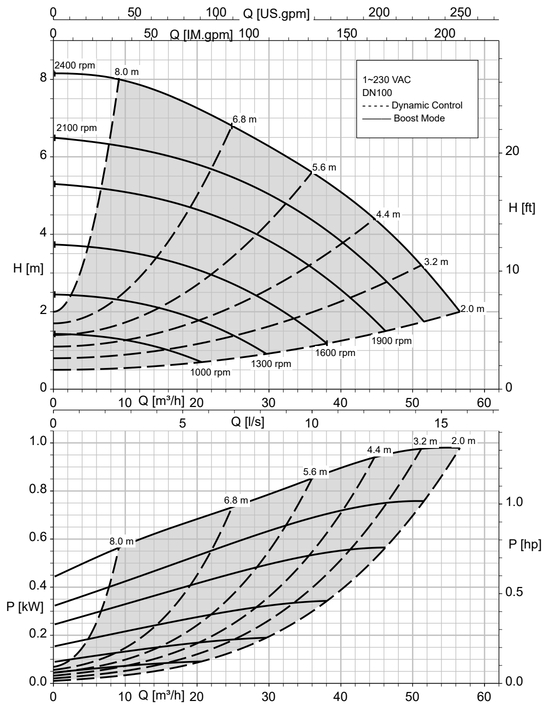
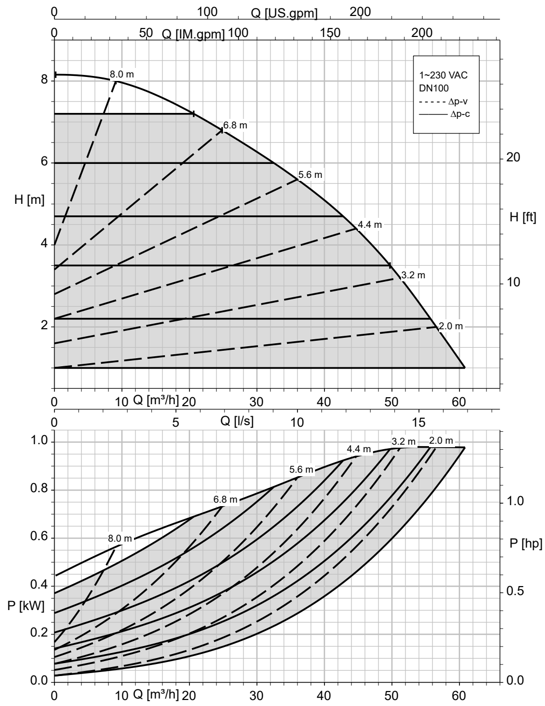
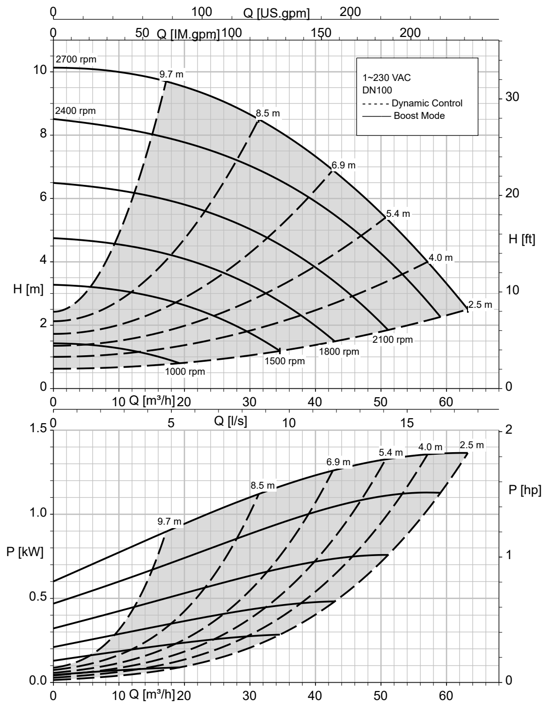
Calio Pro Plus 100-80 Open-loop control, Dynamic Control



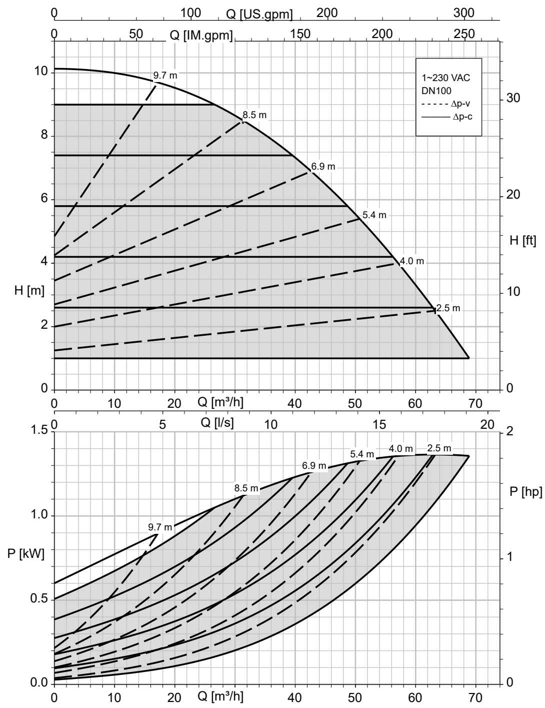
Calio Pro Plus 100-80 Δpv, Δpc



Calio Pro Plus 100-100 Open-loop control, Dynamic Control



Calio Pro Plus 100-100 Δpv, Δpc



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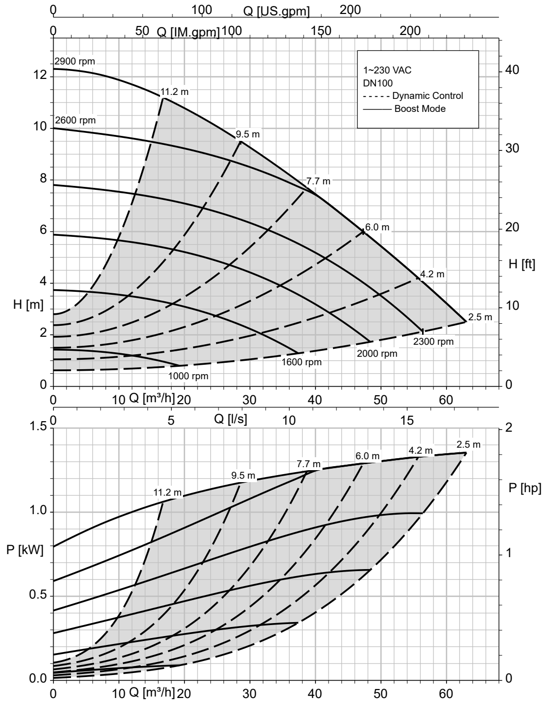
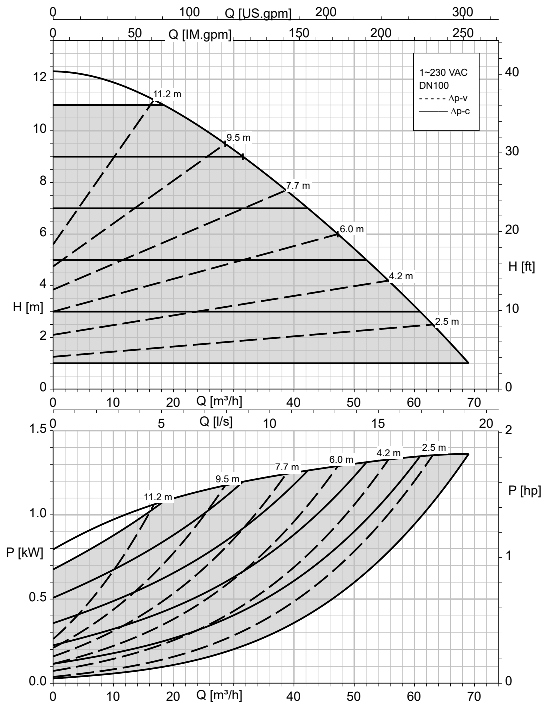
Calio Pro Plus 100-120 Open-loop Control, Dynamic Control



Calio Pro Plus 100-120 Δpv, Δpc



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Dimensions

Pump set dimensions

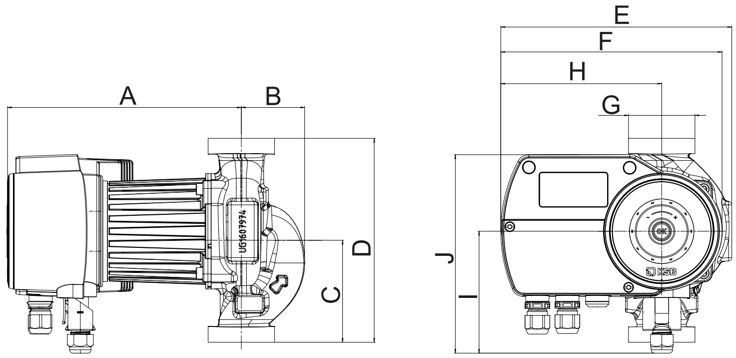

Fig. 6: Screw-ended pump set

Table 10: Dimensions of screw-ended pump set

Size	Connection		A	B	C	D	E	F	H	I	J
	Piping	Pump	[mm]	[mm]	[mm]	[mm]	[mm]	[mm]	[mm]	[mm]	[mm]
25-40	R 1[7]	G 1 1/2	209	56	98	180	205	196	143	108	175
25-60	R 1[7]	G 1 1/2	209	56	98	180	205	196	143	108	175
25-80	R 1[7]	G 1 1/2	209	56	98	180	205	196	143	108	175
25-100	R 1[7]	G 1 1/2	209	56	98	180	205	196	143	108	175
25-120	R 1[7]	G 1 1/2	209	56	98	180	205	196	143	108	175
30-40	R 1 1/4[7]	G 2	209	56	98	180	205	196	143	108	175
30-60	R 1 1/4[7]	G 2	209	56	98	180	205	196	143	108	175
30-80	R 1 1/4[7]	G 2	209	56	98	180	205	196	143	108	175
30-100	R 1 1/4[7]	G 2	209	56	98	180	205	196	143	108	175
30-120	R 1 1/4[7]	G 2	209	56	98	180	205	196	143	108	175
30-140	R 1 1/4[7]	G 2	209	56	98	180	205	196	143	108	175
30-160	R 1 1/4[7]	G 2	209	56	98	180	205	196	143	108	175

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7 Connection using pump pipe unions (accessories)

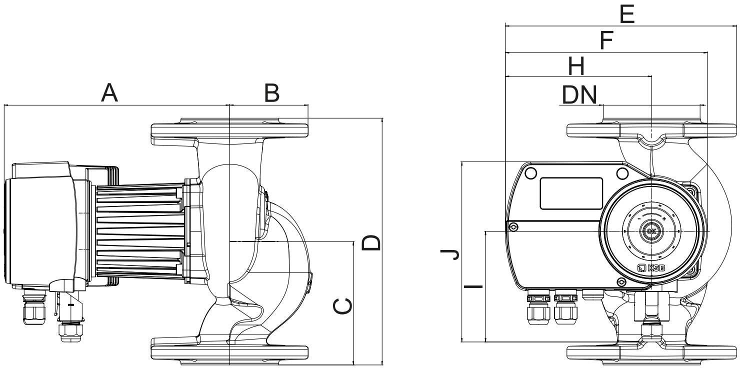


Fig. 7: Flanged pump set (P_1 ≤ 400 W)

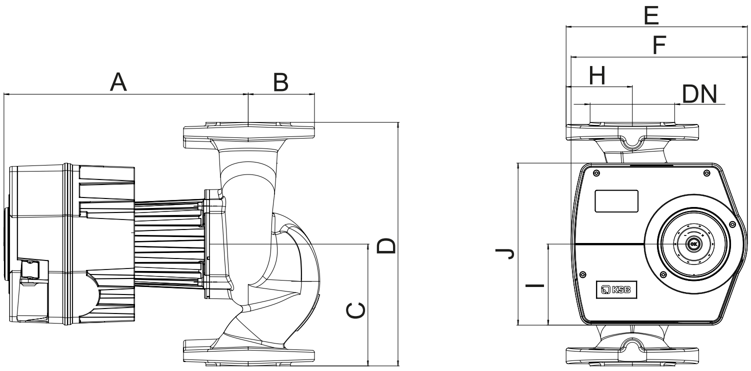


Fig. 8: Flanged pump set (P_1 > 400 W)

Table 11: Dimensions of flanged pump set

Size	Connection		A	B	C	D	E	F	H	I	J
	Piping	Pump	[mm]	[mm]	[mm]	[mm]	[mm]	[mm]	[mm]	[mm]	[mm]
Flanged pump set (P1 ≤ 400 W)											
32-40	DN 32	DN 32	209	65	110	220	213	196	143	108	175
32-60	DN 32	DN 32	209	65	110	220	213	196	143	108	175
32-80	DN 32	DN 32	209	65	110	220	213	196	143	108	175
32-100	DN 32	DN 32	209	65	110	220	213	196	143	108	175
32-120	DN 32	DN 32	209	65	110	220	213	196	143	108	175
32-140	DN 32	DN 32	209	65	110	220	213	196	143	108	175
32-160	DN 32	DN 32	209	65	110	220	213	196	143	108	175
40-40	DN 40	DN 40	217	70	120	220	218	196	143	108	175
40-60	DN 40	DN 40	217	70	120	220	218	196	143	108	175
40-80	DN 40	DN 40	217	70	120	220	218	196	143	108	175
40-100	DN 40	DN 40	217	70	120	220	218	196	143	108	175
50-40	DN 50	DN 50	214	76	120	240	226	196	143	108	175
50-60	DN 50	DN 50	214	76	120	240	226	196	143	108	175
50-80	DN 50	DN 50	214	76	120	240	226	196	143	108	175
65-40	DN 65	DN 65	221	86	170	340	236	196	143	108	175
65-60	DN 65	DN 65	221	86	170	340	236	196	143	108	175
Flanged pump set (P1 > 400 W)											
40-120	DN 40	DN 40	337	73	135	250	247	247	86	114	227
40-150	DN 40	DN 40	337	73	135	250	247	247	86	114	227
40-180	DN 40	DN 40	337	73	135	250	247	247	86	114	227
50-100	DN 50	DN 50	337	78	140	280	247	247	86	114	227
50-120	DN 50	DN 50	337	78	140	280	247	247	86	114	227
50-150	DN 50	DN 50	337	78	140	280	247	247	86	114	227
50-180	DN 50	DN 50	337	78	140	280	247	247	86	114	227
65-80	DN 65	DN 65	341	93	170	340	254	247	93	114	227
65-100	DN 65	DN 65	341	93	170	340	254	247	93	114	227
65-120	DN 65	DN 65	341	93	170	340	254	247	93	114	227
80-40	DN 80	DN 80	341	103	170	360	261	247	100	114	227

Size	Connection		A	B	C	D	E	F	H	I	J
	Piping	**Pump**	**[mm]**	**[mm]**	**[mm]**	**[mm]**	**[mm]**	**[mm]**	**[mm]**	**[mm]**	**[mm]**
80-60	DN 80	DN 80	341	103	170	360	261	247	100	114	227
80-80	DN 80	DN 80	341	103	170	360	261	247	100	114	227
80-100	DN 80	DN 80	341	103	170	360	261	247	100	114	227
80-120	DN 80	DN 80	341	103	170	360	261	247	100	114	227
80-160	DN 80	DN 80	341	103	170	360	261	247	100	114	227
100-40	DN 100	DN 100	343	120	210	450	271	247	110	114	227
100-60	DN 100	DN 100	343	120	210	450	271	247	110	114	227
100-80	DN 100	DN 100	343	120	210	450	271	247	110	114	227
100-100	DN 100	DN 100	343	120	210	450	271	247	110	114	227
100-120	DN 100	DN 100	343	120	210	450	271	247	110	114	227

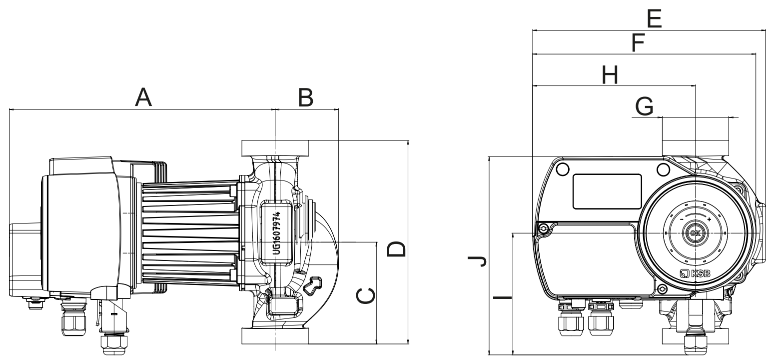


Fig. 9: Screw-ended pump set with field bus module (accessory)

Table 12: Dimensions of screw-ended pump set with field bus module (accessory)

Size	Connection		A	B	C	D	E	F	H	I	J
	Piping	**Pump**	**[mm]**	**[mm]**	**[mm]**	**[mm]**	**[mm]**	**[mm]**	**[mm]**	**[mm]**	**[mm]**
25-40	R 1[7)]	G 1 1/2	236	56	98	180	207	198	145	108	175
25-60	R 1[7)]	G 1 1/2	236	56	98	180	207	198	145	108	175
25-80	R 1[7)]	G 1 1/2	236	56	98	180	207	198	143	108	175
25-100	R 1[7)]	G 1 1/2	236	56	98	180	207	198	145	108	175
25-120	R 1[7)]	G 1 1/2	236	56	98	180	207	198	145	108	175
30-40	R 1 1/4[7)]	G 2	236	56	98	180	207	198	145	108	175
30-60	R 1 1/4[7)]	G 2	236	56	98	180	207	198	145	108	175
30-80	R 1 1/4[7)]	G 2	236	56	98	180	207	198	145	108	175
30-100	R 1 1/4[7)]	G 2	236	56	98	180	207	198	145	108	175
30-120	R 1 1/4[7)]	G 2	236	56	98	180	207	198	145	108	175
30-140	R 1 1/4[7)]	G 2	236	56	98	180	207	198	145	108	175
30-160	R 1 1/4[7)]	G 2	236	56	98	180	207	198	145	108	175

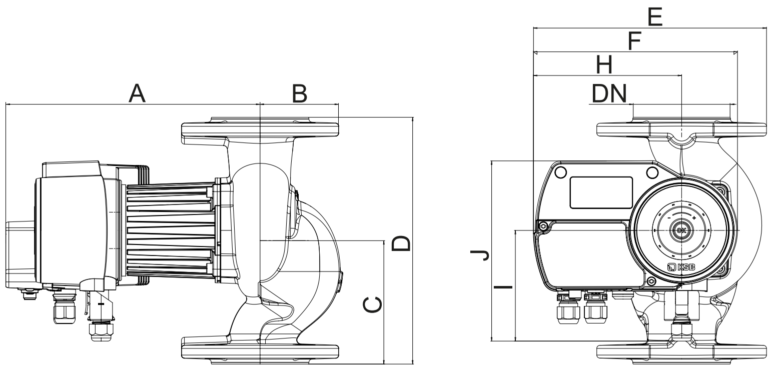


Fig. 10: Flanged pump set with field bus module (accessory)

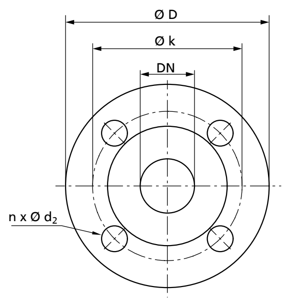
Table 13: Dimensions of flanged pump set with field bus module (accessory)

Size	Connection		A	B	C	D	E	F	H	I	J
	Piping	Pump	[mm]	[mm]	[mm]	[mm]	[mm]	[mm]	[mm]	[mm]	[mm]
32-40	DN 32	DN 32	236	65	110	220	215	198	145	108	175
32-60	DN 32	DN 32	236	65	110	220	215	198	145	108	175
32-80	DN 32	DN 32	236	65	110	220	215	198	145	108	175
32-100	DN 32	DN 32	236	65	110	220	215	198	145	108	175
32-120	DN 32	DN 32	236	65	110	220	215	198	145	108	175
32-140	DN 32	DN 32	236	65	110	220	215	198	145	108	175
32-160	DN 32	DN 32	236	65	110	220	215	198	145	108	175
40-40	DN 40	DN 40	244	70	120	220	220	198	145	108	175
40-60	DN 40	DN 40	244	70	120	220	220	198	145	108	175
40-80	DN 40	DN 40	244	70	120	220	220	198	145	108	175
40-100	DN 40	DN 40	244	70	120	220	220	198	145	108	175
50-40	DN 50	DN 50	241	76	120	240	228	198	145	108	175
50-60	DN 50	DN 50	241	76	120	240	228	198	145	108	175
50-80	DN 50	DN 50	241	76	120	240	228	198	145	108	175
65-40	DN 65	DN 65	248	86	170	340	238	198	145	108	175
65-60	DN 65	DN 65	248	86	170	340	238	198	145	108	175

Flange dimensions

Table 14: Flange dimensions

Size	PN 6			PN 10, PN 16			Outline drawing
	Ø D	Ø k	n × Ø d_2	Ø D	Ø k	n × Ø d_2	
	[mm]	[mm]	[mm]	[mm]	[mm]	[mm]	
DN 32	120	90	4 × Ø 14	140	100	4 × Ø 19	
DN 40	130	100	4 × Ø 14	150	110	4 × Ø 19	
DN 50	140	110	4 × Ø 14	165	125	4 × Ø 19	
DN 65	160	130	4 × Ø 14	185	145	4 × Ø 19	
DN 80	190	150	4 × Ø 19	200	160	8 × Ø 19	
DN 100	210	170	4 × Ø 19	220	180	8 × Ø 19	



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Installation information

Permissible installation positions

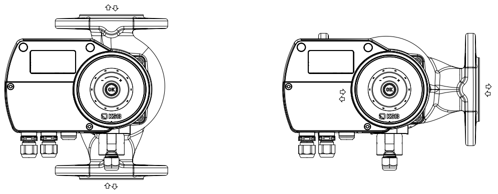

Fig. 11: Permissible installation positions

Scope of supply

Depending on the model, the following items are included in the scope of supply:

- Pump set
- Sealing elements
- Electrical plug-type connector (only for pump sets with $P_1 \leq$ 400 W)
- Washers (only for flanged pump sets of DN 32 to DN 65)
- Installation/operating manual
- Two-piece thermal insulation shell

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Accessories

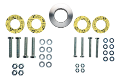
Pipe unions

Table 15: Pipe unions

	Description	Mat. No.	[kg]
	2 pipe unions with G 1 1/2 union nut and insert with Rp 3/4 internal thread, steel for pumps with G 1 1/2 external thread / Rp 3/4 pipe connection	19075560	0.2
	2 pipe unions with G 1 1/2 union nut and insert with Rp 1 internal thread, steel for pumps with G 1 1/2 external thread / Rp 1 pipe connection	19075561	0.2
	2 pipe unions with G 2 union nut and insert with Rp 1 1/4 internal thread, steel for pumps with G 2 external thread / Rp 1 1/4 pipe connection	19075562	0.2

Spacers (flange)

Table 16: Spacers (flange)

	Description	Connection	PN	Length	Mat. No.	[kg]
				[mm]		
	Spacer F16	DN 40	6/10/16	30	19075991	2
	Spacer F0	DN 40	6/10/16	70	19075566	2
	Spacer F1	DN 50	6/10/16	10	19075567	2
	Spacer F2	DN 50	6/10/16	20	19075568	2
	Spacer F3	DN 50	6/10/16	50	19075569	2
	Spacer F4	DN 50	6/10/16	60	19075570	2
	Spacer F5	DN 65	6/10/16	10	19075571	2
	Spacer F6	DN 65	6/10/16	25	19075572	2
	Spacer F7	DN 65	6/10/16	30	19075573	2
	Spacer F8	DN 80	6/10/16	10	19075574	2
	Spacer F9	DN 80	6/10/16	15	19075575	2
	Spacer F10	DN 80	6/10/16	20	19075576	2
	Spacer F11	DN 80	6/10/16	25	19075577	2
	Spacer F12	DN 80	6/10/16	30	19075578	2
	Spacer F13	DN 80	6/10/16	40	19075579	2
	Spacer F14	DN 80	6/10/16	50	19075580	2
	Spacer F15	DN 80	6/10/16	80	19075581	2

Electrical accessories

Table 17: Electrical accessories

	Description	Connection	Electric cable length	Measuring range	Mat. No.	[kg]
			[m]	[°C]		
	Resistance thermometer (cable sensor) Pt1000, 6 × 50 × 3000	DN 25 - 80	3,00	-50 - +180	05360148	0.1
	Resistance thermometer (cable sensor) Pt1000, 6 × 50 × 5000	DN 25 - 80	5,00		05360149	0.18
	Resistance thermometer (cable sensor) Pt1000, 6 × 50 × 10000	DN 25 - 80	10,00		05360230	0.29
	Resistance thermometer (cable sensor) Pt1000, 6 × 100 × 3000	100 - 200	3,00		05360231	0.12
	Resistance thermometer (cable sensor) Pt1000, 6 × 100 × 5000	100 - 200	5,00		05360232	0.19
	Resistance thermometer (cable sensor) Pt1000, 6 × 100 × 10000	DN 100 - 200	10,00		05360233	0.35
	Protective well G 1/2, hex. WAF 22 × 50	DN 25 - 80	-	-	05360237	0.08
	Protective well G 1/2, hex. WAF 22 × 100	100 - 200	-	-	05360238	0.09

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